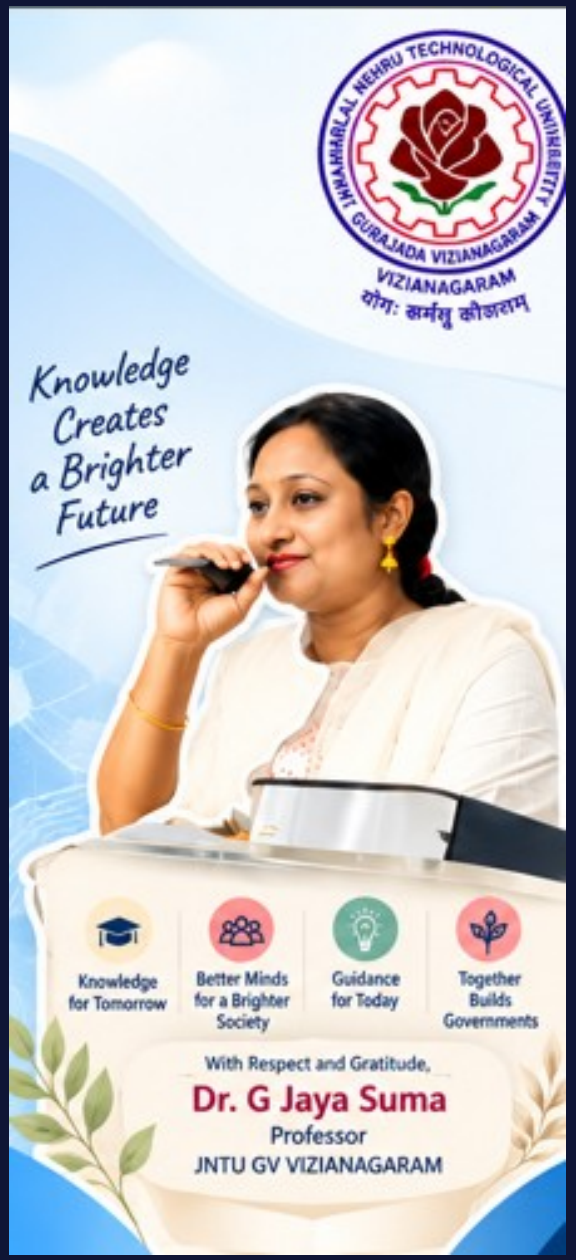


JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY
GURAJADA VIZIANAGARAM

TECHNOLOGY FOR A BETTER TOMORROW

Learn
Innovate
Create
Lead

VIDEO LECTURE SERIES · QUANTUM APPLICATIONS

Introduction to Unit V

Where quantum algorithms meet real problems

One shared pipeline, two hardware eras, and a roadmap of the application domains ahead.

10 MIN

Encode | Evolve | Measure | Post-process

1 What we’ll cover in 10 minutes

Four short segments: the shared recipe, the two hardware eras, the roadmap, and a wrap-up.

1 One pipeline for every application

Encode, evolve, measure, post-process, shown across five domains.

4 min

2 NISQ versus fault-tolerant era

What runs on today’s noisy devices and what must wait for error correction.

3 min

3 Roadmap of Unit V

The domains we will study, and where each sits on the timeline.

2 min

4 Wrap-up and how to study

One takeaway and five questions to ask of every topic.

1 min

1 Recap: from algorithms to applications

We know the building blocks; now we ask what real problems they can serve.

In simple words

So far we studied algorithms one at a time: search, Fourier transforms, phase estimation. An application is an algorithm wrapped around a real problem.

Algorithm *+ data* **Application**

The hard part is rarely the algorithm itself. It is moving real data in and a usable answer out.

A

Algorithms

Grover, QFT, phase estimation: reusable quantum building blocks.

P

Problems

Molecules, schedules, portfolios, images, keys: real tasks with real data.

?

The gap

How does a real problem become a circuit, and a measurement a real answer?

1 One pipeline for every application

Whatever the domain, the workflow has the same four stages.

1

Encode

Turn the problem into qubits: states, Hamiltonians or cost functions.

Classical

→

2

Evolve

Apply a circuit of gates. Some gate angles may be tunable parameters.

Quantum

→

3

Measure

Sample the register many times. Each shot returns just one bitstring.

Quantum

→

4

Post-process

Classical step: average, optimise, decode, or verify the answer.

Classical

Key idea: only the problem-specific stages (encode and post-process) change from domain to domain. Evolve and measure reuse the same quantum toolkit, and in near-term methods stage 4 loops back to stage 2.

1 Same pipeline, five domains

Each application fills the four stages with its own content.

Application	Encode	Evolve	Measure	Post-process
Chemistry	Molecule to qubit Hamiltonian	Ansatz circuit or phase estimation	Energy terms, one Pauli at a time	Sum terms, minimise energy
Optimisation	Cost function to Ising problem	Alternating QAOA layers	Bitstring samples	Keep the best cut, refine
Machine learning	Data to a quantum feature map	Parametrised circuit	Class or kernel estimates	Classical optimiser trains weights
Finance	Distribution to amplitudes	Amplitude estimation	Counts of a marker qubit	Convert to a price or risk number
Cryptography	Number to modular arithmetic	Order finding with QFT	Phase bitstring	Continued fractions give factors

1 Worked example: MaxCut optimisation

Follow one concrete problem through all four stages.

Example: split a graph into two groups

graph → Ising cost → QAOA → bitstrings → best cut

The goal is to cut as many edges as possible. Each node becomes a qubit, and a bitstring names the two groups.

2
Classical stages
encode, post-process

2
Quantum stages
evolve, measure

+

Encode

Every edge becomes a cost term, and every node a qubit.

~

Evolve and measure

Alternate cost and mixer layers, then sample many bitstrings.

✓

Post-process

Pick the best cut found, tune the angles, and repeat.

2 Two eras: NISQ and fault-tolerant

The same pipeline runs on very different hardware assumptions.

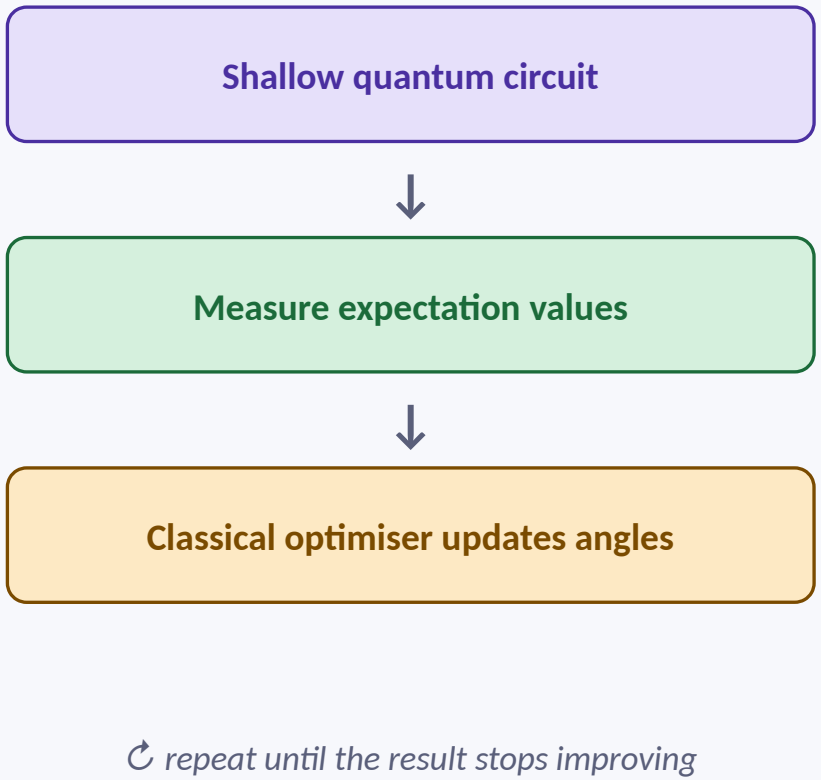
Aspect	NISQ (today)	Fault-tolerant (future)
Qubits	Tens to hundreds, noisy	Many logical qubits built from many physical ones
Errors	Uncorrected; they cap circuit depth	Detected and corrected while running
Circuit depth	Shallow circuits, short runs	Deep circuits with millions of gates
Algorithm style	Hybrid, variational, heuristic	Provable algorithms such as Shor and phase estimation
Advantage	Still an open question	Proven speedups for some tasks

Rule of thumb: the deeper the circuit an application needs, the later the era it belongs to.

2 Near-term (NISQ) applications

Short circuits inside a classical loop, built to tolerate noise.

The hybrid loop



V

Variational chemistry

VQE estimates molecular ground-state energies on small molecules.

Q

Optimisation

QAOA and relatives explore routing, scheduling and graph problems.

M

Quantum machine learning

Parametrised circuits and quantum kernels classify or cluster data.

2 Fault-tolerant era applications

Deep, provable algorithms that need error-corrected qubits.

S

Factoring and cryptanalysis

Shor’s algorithm breaks RSA-style keys, driving post-quantum cryptography.

C

Precise chemistry and materials

Phase estimation targets chemical accuracy beyond classical reach.

G

Search and amplitude estimation

Quadratic speedups for search, sampling and Monte Carlo methods.

H

Large-scale simulation

Hamiltonian simulation of dynamics for materials and physics.

Why wait? These algorithms need thousands of logical qubits and deep circuits, far beyond today’s noisy devices.

3 Roadmap of Unit V

Six application domains, each told through the same four stages.

1 — NISQ + FT

Chemistry and materials

Molecular energies and reaction modelling.

2 — Mostly FT

Quantum simulation

Dynamics of spins, lattices and fields.

3 — NISQ

Optimisation

Routing, scheduling and logistics.

4 — NISQ

Machine learning

Quantum kernels and trainable circuits.

5 — NISQ + FT

Finance

Risk estimation and portfolio problems.

6 — FT

Cryptography and security

Shor’s threat and quantum-safe defences.

3 Where each domain sits on the timeline

Capability follows hardware: more qubits, lower errors, deeper circuits.

TODAY — NEXT — LATER

1 — 2 — 3

NISQ

Optimisation, machine learning and small-molecule chemistry as hybrid experiments.

Early fault tolerance

A few error-corrected qubits improve chemistry, simulation and finance estimates.

Full fault tolerance

Factoring, large simulation and precise materials design.

Remember: the pipeline stays the same across eras. Better hardware lets us run deeper evolve stages.

ONE-LINE TAKEAWAY

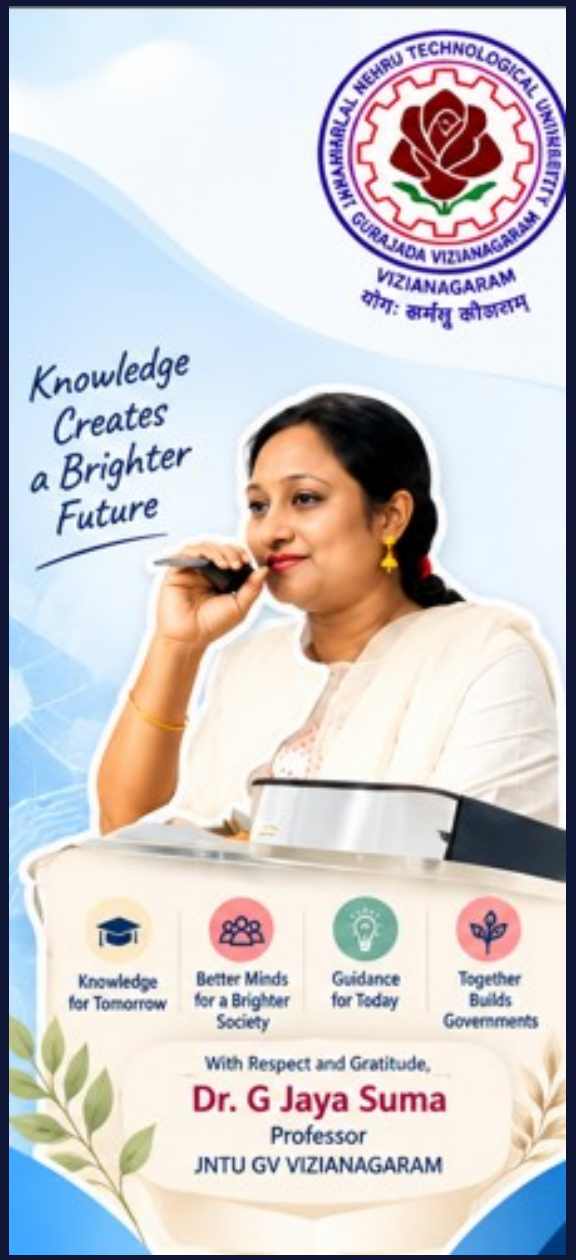
Every application runs encode, evolve, measure, post-process. What differs is the problem, and whether it needs noisy or error-corrected hardware.

Encode | Evolve | Measure | Post-process

Ask these five questions in every topic

What is encoded? · What evolves? · What is measured? · How is it post-processed? · NISQ or fault-tolerant?

Next: the first application domain, quantum chemistry and materials.



JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY
GURAJADA VIZIANAGARAM

TECHNOLOGY FOR A BETTER TOMORROW

Learn
Innovate
Create
Lead

VIDEO LECTURE SERIES · QUANTUM APPLICATIONS

Quantum Cryptography and Key Distribution

Keys protected by the laws of physics: BB84, E91, and how real QKD networks work.

55 MIN

No-cloning | BB84 | Eavesdropping | E91

1 What we'll cover in 55 minutes

Six segments: from the key-exchange problem to a hands-on eavesdropping analysis.

1	**The key-exchange problem and the physics** Classical key exchange, no-cloning, measurement disturbance.	10 min
2	**BB84 and a worked run** Bases, encoding, sifting, error estimation, privacy amplification.	15 min
3	**Eavesdropping and the 25% signature** Intercept-and-resend and how Alice and Bob detect it.	8 min
4	**E91 and practical QKD** Entanglement-based QKD, fibre, satellites, trusted nodes, deployments.	10 min
5	**Demo: simulating BB84** Run the protocol with and without an eavesdropper.	6 min
6	**Exercise: analyse an attack** Work out whether Eve was present, from the numbers.	6 min

1 Classical key exchange

Two parties who have never met must still agree on a shared secret key.

In simple words

Fast ciphers such as AES need both sides to hold the same secret key. Key exchange is how they get it over a channel anyone can listen to.



The secret appears only at the two ends, even though everything sent was public.

Diffie-Hellman

D

Each side combines a public value with a private exponent. Security rests on the discrete logarithm problem.

RSA key transport

R

One side encrypts a fresh key with the other's public key. Security rests on factoring large numbers.

The catch

!

Both rely on maths problems that are believed hard, not proven hard.

1 The problem quantum solves

A future quantum computer breaks the maths behind today's key exchange.

	Classical key exchange	Quantum key distribution
Security rests on	Hard maths problems	Laws of quantum physics
Quantum computers	Shor's algorithm breaks it	Do not help the attacker
Recorded traffic	Can be decrypted later	Key was never sent as data
Eavesdropping	Leaves no trace	Disturbs the signal, so is detectable
Security claim	Computational (assumed)	Information-theoretic (provable)

Be precise: QKD only distributes a key. The data is then encrypted with it, ideally by a one-time pad, and both sides must authenticate the classical channel.

1 Principle 1: the no-cloning theorem

An unknown quantum state cannot be copied perfectly.

The statement

No unitary U gives $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ for every $|\psi\rangle$

Why: unitaries preserve inner products. Copying would turn $\langle\psi|\varphi\rangle$ into $\langle\psi|\varphi\rangle^2$, which only works if the states are identical or orthogonal.

Classical: copy a bit, read it, forward it, nobody notices.
Quantum: an unknown qubit cannot be duplicated, so Eve cannot keep a copy.

1

Perfect copying is ruled out

Even a quantum copier trades accuracy for disturbance.

2

Orthogonal states can be cloned

Only if you already know the basis, which Eve does not.

3

Eve's dilemma

Any information she takes costs a disturbance Bob can see.

1 Principle 2: measurement disturbs the state

Measuring in the wrong basis gives a random answer and changes the qubit.

Qubit prepared as	Measured in Z basis	Measured in X basis
$\|0\rangle$ (Z basis, bit 0)	0 with certainty	0 or 1, 50% each
$\|1\rangle$ (Z basis, bit 1)	1 with certainty	0 or 1, 50% each
$\|+\rangle$ (X basis, bit 0)	0 or 1, 50% each	0 with certainty
$\|-\rangle$ (X basis, bit 1)	0 or 1, 50% each	1 with certainty

✓

Right basis

The result is certain and the state is left unchanged.

✗

Wrong basis

A random result, and the original state is lost.

?

Why Eve is stuck

She does not know the basis, so she guesses wrong half the time.

2 BB84: bases and encoding

Alice hides each bit in one of two bases that cannot be told apart without knowing which.

Basis	Bit 0	Bit 1	Photon polarisation
Z (rectilinear)	$\|0\rangle$	$\|1\rangle$	0° and 90°
X (diagonal)	$\|+\rangle$	$\|-\rangle$	45° and 135°

A

Alice

Picks a random bit and a random basis for every qubit, then sends the resulting state.

B

Bob

Picks his own random basis for every qubit and measures. He does not know Alice's choice.

K

The trick

Bases are announced after the measurement. Only then can both sides tell which results to keep.

2 BB84 at a glance: five stages

Two quantum steps, then three classical steps over an authenticated channel.

1 — Encode

Alice sends random bits in random bases.

Quantum

→

2 — Measure

Bob measures each qubit in a random basis.

Quantum

→

3 — Sift

Announce bases; keep only the matching positions.

Classical

→

4 — Estimate error

Compare a sample of bits to find the error rate.

Classical

→

5 — Privacy amplif.

Hash the remaining key to a shorter secret one.

Classical

Key idea: this is the pipeline from the Unit V introduction. Encode and measure are quantum; sifting, error estimation and privacy amplification are classical post-processing.

2 Worked run, part 1: send, measure, sift

Ten qubits between Alice and Bob, no eavesdropper.

#	1	2	3	4	5	6	7	8	9	10
Alice bit	0	1	1	0	1	0	0	1	1	0
Alice basis	Z	X	X	Z	X	Z	X	Z	Z	X
Bob basis	Z	Z	X	X	X	Z	Z	Z	X	X
Bob result	0	0	1	1	1	0	1	1	0	0
Bases match?	✓	×	✓	×	✓	✓	×	✓	×	✓
Key bit	0	–	1	–	1	0	–	1	–	0

Green columns: bases agree, so Bob's result equals Alice's bit. **Red columns:** bases differ, so Bob's result is a coin flip and the position is discarded.

Sifted key: 011010 (6 of 10 positions survive, about half on average).

2 Worked run, part 2: from sifted key to secret key

Follow the same ten qubits through the three classical steps.

3 Sift

Keep matching bases only.

`0 1 1 0 1 0`

→

4 Estimate error

Publicly compare bits 2 and 5 of the sifted key (both 1): 0 errors, QBER = 0%.

`0 1 · 0 · 0`

→

5 Privacy amplification

Hash pairs with XOR: 0 XOR 1 and 0 XOR 0. A shorter key Eve knows nothing about.

`1 0`

Toy numbers: real runs use thousands of qubits and a universal hash function; ten qubits only show the mechanics.

2 Error estimation and privacy amplification

Turning a noisy, partly exposed raw key into a short secret one.

E

Error correction

Alice and Bob reconcile their strings over the public channel, for example with parity checks. It fixes errors but leaks some bits.

Q

Error estimate (QBER)

QBER = errors ÷ sampled bits. If it exceeds about 11%, no secret key can be guaranteed, so they abort.

P

Privacy amplification

A universal hash shrinks the key by the amount Eve might know, leaving a shorter but secret key.

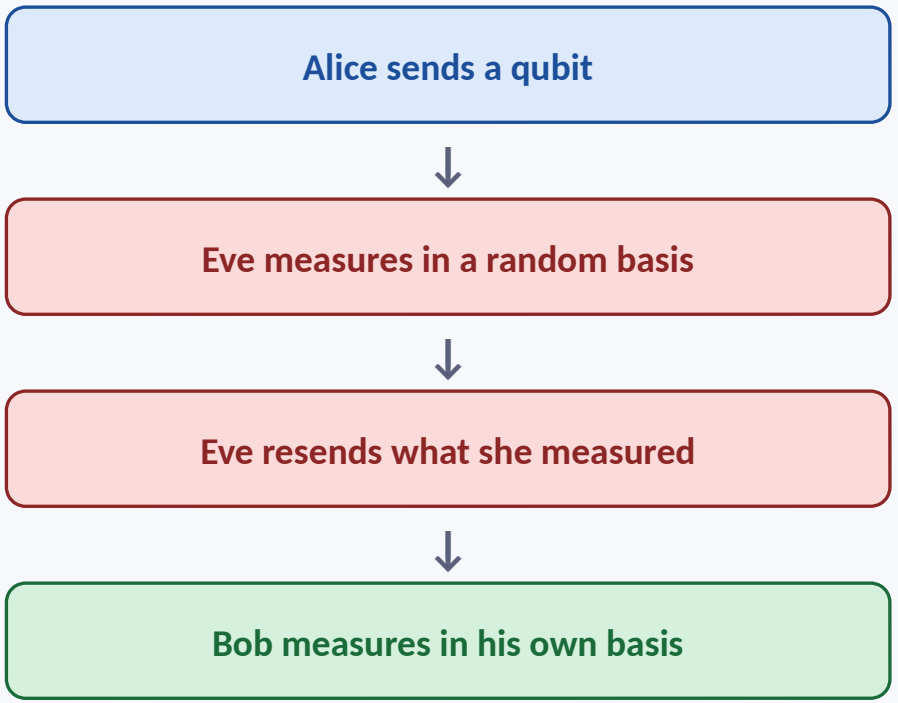
raw key → correct errors → measure QBER → hash to final key

QBER below the threshold: a secret key can be extracted. Above it: abort and start again.

3 Intercept-and-resend attack

Eve measures each qubit and forwards a replacement, hoping nobody notices.

What Eve does



Every qubit she touches is replaced by her own guess.

Right basis (50%)

✓

Eve reads the true bit and resends the exact state. Nobody can tell.

Wrong basis (50%)

✗

Eve gets a random result and resends the wrong state, disturbing it.

No-cloning bites

!

She cannot keep a copy and forward the original. She must disturb it.

3 The 25% error signature

Eve's guess errors leave a fingerprint Alice and Bob can measure.

50%

Eve picks the wrong basis

× 50%

Bob then reads a wrong bit

= 25%

Error rate on the sifted key

Chance Eve goes undetected after Alice and Bob compare n sampled bits: $(3/4)^n$

20 bits

99.7%

detection probability

50 bits

99.99994%

detection probability

25 bits

≥ 99.9%

the minimum sample for 99.9%

Takeaway: a full intercept-and-resend attack pushes the error rate to 25%. Natural noise is usually just a few percent, so the jump is visible.

3 Detecting Eve in practice

Noise and eavesdropping look the same, so every error is treated as an attack.

A partial attack

$$\text{QBER} \approx 25\% \times f$$

f is the fraction of qubits Eve intercepts. Attack fewer qubits and you learn less, but you also leave a smaller trace.

If QBER stays below about 11%, privacy amplification can still remove what Eve might know. Above that, they abort.

1

Measure the QBER

Sample part of the sifted key and publicly compare it.

2

Assume the worst

Count every error as caused by Eve, even natural noise.

3

Decide

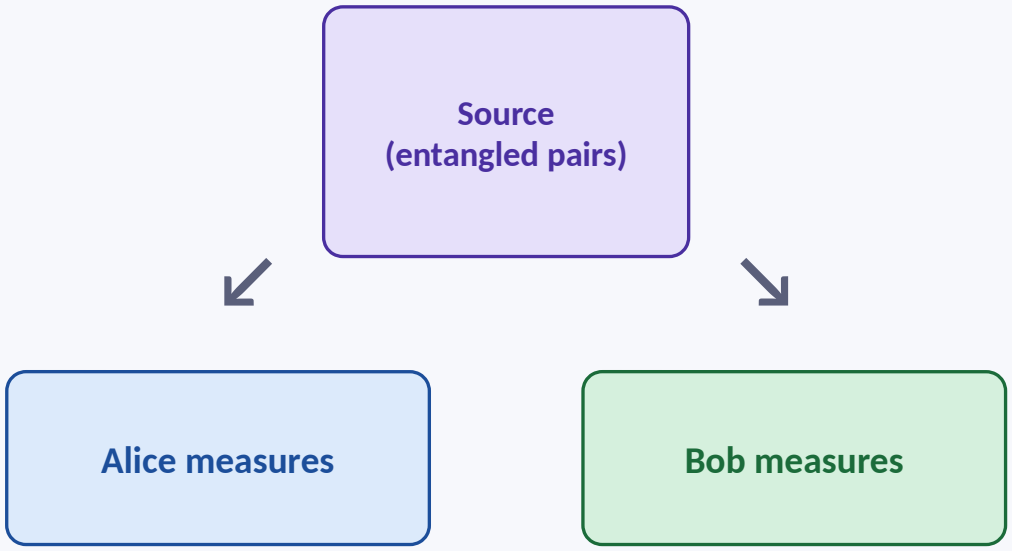
Below threshold: shrink the key and use it. Above: abort and retry.

4

E91: entanglement-based QKD

Ekert (1991): the key comes from shared entangled pairs, not from sent bits.

The setup



Eve cannot know the outcome in advance: it does not exist until the measurement.

1

Distribute pairs

A source sends one qubit of each entangled pair to Alice and one to Bob.

2

Measure at random

Each picks a measurement angle at random and records the result.

3

Compare angles

Matching angles give correlated key bits. The other angles go into a Bell test.

4 E91 versus BB84

Different ingredients, same goal: a shared secret key and a way to detect Eve.

	BB84	E91
Ingredient	Single qubits sent from Alice to Bob	Entangled pairs from a source
Key comes from	Alice’s bits in matching bases	Correlated outcomes at matching angles
Eve test	Error rate (QBER) on a sample	Bell (CHSH) value S on the rest
Secure if	QBER below about 11%	S well above 2 (maximum $2\sqrt{2} \approx 2.83$)
Strength	Simple and widely deployed	Security can rest on the measured violation, not on trusting the devices

Equivalence: the two protocols are closely related; BB84 can be seen as an entanglement-based scheme with the source on Alice’s side.

4 Practical QKD: fibre links

Photons travel well in fibre, but they cannot be amplified.

0.2 dB

loss per km in telecom fibre (1550 nm)

No

amplifiers: they would need to copy the qubit

100s km

range reached in lab with twin-field QKD

L

Loss limits the rate

Each extra distance step loses photons, so key rate drops exponentially with length.

D

Detectors matter

Single-photon detectors and their noise set the maximum distance in practice.

T

Twin-field and beyond

New schemes interfere signals from both ends to reach much further.

4 Satellites and trusted nodes

Two ways to go beyond the reach of a single fibre link.

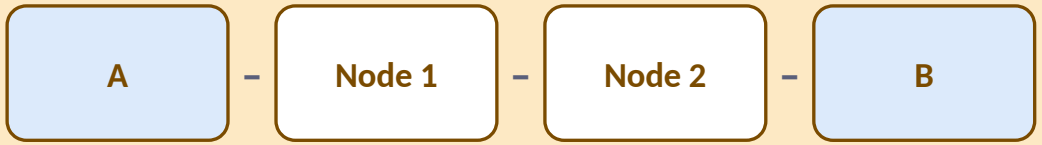
S Satellite QKD

Free-space loss grows slowly with distance, so a satellite can link ground stations thousands of km apart.

Micius (China, launched 2016) distributed entanglement over about 1,200 km and enabled intercontinental key exchange between China and Austria.

Limits: weather, daylight and a short pass time over each station.

T Trusted-node chains



Each link makes its own QKD key; the nodes relay the end-to-end key by encrypting it hop by hop.

Catch: every node sees the key in plaintext, so every node must be physically secured and trusted.

Future quantum repeaters could remove this need.

4 Real deployments

From laboratory demonstrations to national networks.

2008–10 **City networks**

SECOQC in Vienna, the Tokyo QKD network and SwissQuantum in Geneva showed multi-user QKD.

2016–17 **Satellite**

Micius: space-to-ground QKD and entanglement distribution across continents.

2017 **Backbone**

Beijing–Shanghai trusted-node backbone of about 2,000 km, linked to the satellite.

Now **Networks and products**

EU's EuroQCI plans, plus commercial systems from companies such as ID Quantique and Toshiba.

Note: deployment details change fast, so treat these as examples and check recent sources for current status.

4 Limits of QKD and the alternatives

A fair assessment: strong physics, demanding engineering.

Device attacks

!

Real detectors and sources leak. Blinding and side-channel attacks target hardware, not the theory.

Cost and reach

$

Special hardware, no amplifiers, and trusted nodes for long distance.

Authentication needed

A

The classical channel must be authenticated, using a pre-shared key or a signature scheme.

Alternative: post-quantum cryptography

New classical algorithms (lattice-based and others) believed to resist quantum attacks.

Run on ordinary hardware and ordinary networks.

Security is computational, not information-theoretic.

Many national agencies prefer it for general use. Some systems combine both.

5 Demo: simulating BB84 with an eavesdropper

Plain Python, no quantum library needed: the model tracks each qubit's bit and basis.

```
for bit, a, b in zip(bits, a_bases, b_bases):
    state_bit, state_basis = bit, a
    if random() < eve_frac:          # Eve attacks
        e = randint(0, 1)            # her random basis
        if e != state_basis:
            state_bit = randint(0, 1)
        state_basis = e              # she resends
    if b != state_basis:             # Bob measures
        state_bit = randint(0, 1)
    bob_bits.append(state_bit)

keep = [i for i in range(n) if a_bases[i]==b_bases[i]]
test = random.sample(keep, len(keep)//5)
errors = sum(bits[i] != bob_bits[i] for i in test)
qber = errors / len(test)

print(f"QBER = {qber:.1%}")
```

1 Run with no Eve

Predict: QBER should be 0%, since the channel is perfect.

2 Run with Eve at 100%

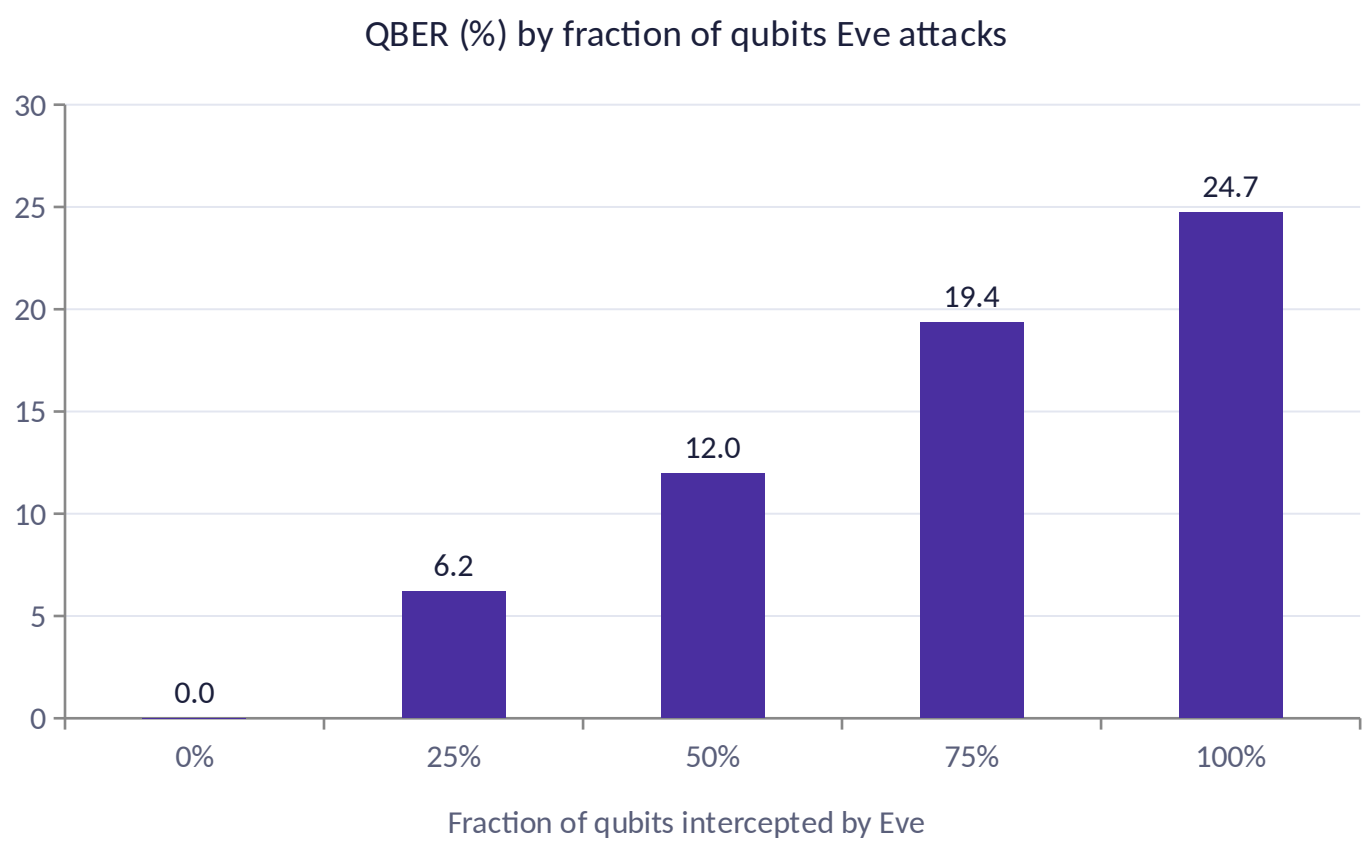
Predict: about 25% of the sifted sample disagrees.

3 Vary eve_frac

Does QBER track 25% × f? Try 0.25, 0.5 and 0.75.

5 Demo results: QBER against attack fraction

Mean of 10 runs, 20,000 qubits each; the error rate tracks 25% × f.



0

No Eve: 0%

A perfect channel shows no errors at all.

~

Matches 25% × f

Measured values sit close to the prediction for each f.

!

About 11% is the limit

At 50% attack the QBER is already near the abort threshold.

6 Exercise: analyse an eavesdropping scenario

Alice and Bob finish a BB84 run and compare their notes.

The data

Quantity	Value
Qubits sent	2,000
Sifted key length	≈ 1,000
Bits compared	200
Errors found	26

Assume natural channel noise is negligible and that any attack is intercept-and-resend.

a **Compute the QBER**

What error rate did Alice and Bob measure?

b **Decide**

Compare with the ~11% threshold: should they keep the key or abort?

c **Estimate Eve's reach**

If every error is Eve's, what fraction f of qubits did she intercept?

d **Plan a test**

How many compared bits give at least 99.9% chance of catching a full attack?

6 Exercise: worked solution

Four short calculations lead to one clear decision.

a **QBER = 26 ÷ 200 = 13%**

Errors divided by bits compared.

b **13% > 11%: abort**

The error rate is above the threshold, so no secret key can be guaranteed. Discard the run.

c **f ≈ 13 ÷ 25 ≈ 52%**

Using QBER ≈ 25% × f, Eve intercepted roughly half of the qubits.

d **n ≥ 25 bits**

Solve $1 - (3/4)^n \geq 0.999$: $n \geq \ln(0.001) \div \ln(0.75) \approx 24.01$, so n = 25.

Reflect: the 200-bit sample was far more than the 25 needed, so Eve's attack was detected with near certainty.

ONE-LINE TAKEAWAY

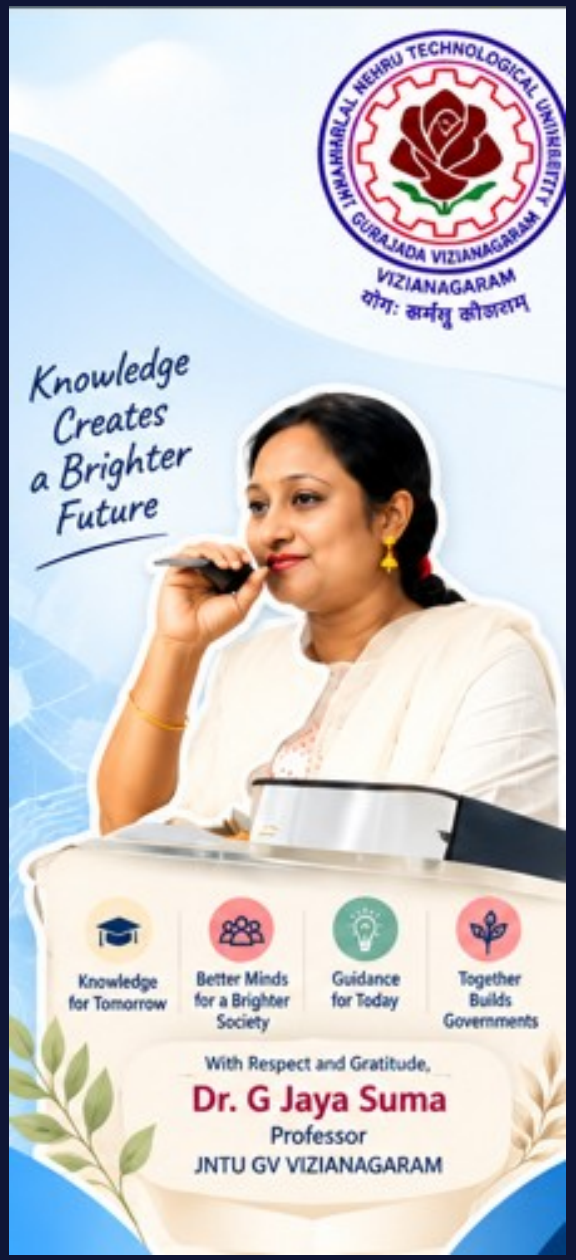
QKD turns no-cloning and measurement disturbance into a tripwire: any eavesdropper leaves a detectable error, and a secret key is kept only when the error is low.

No-cloning | BB84 sifting | 25% signature | E91: Bell test

Try it yourself

Add channel noise to the simulation, then try a different attack, or implement the E91 Bell test.

Remember: QKD distributes keys; it does not encrypt data, and the classical channel must be authenticated.



JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY
GURAJADA VIZIANAGARAM

TECHNOLOGY FOR A BETTER TOMORROW

Learn
Innovate
Create
Lead

VIDEO LECTURE SERIES · QUANTUM ALGORITHMS

Post-Quantum Cryptography

Preparing today's keys and certificates for tomorrow's quantum computers: what breaks, what replaces it, and how to migrate.

40 MIN

Shor threat | Lattices | NIST standards | Migration

A

What we'll cover in 40 minutes

Five segments: from the quantum threat to a hands-on comparison of security assumptions.

1

The quantum threat

Shor, "harvest now, decrypt later", what breaks, and why symmetric crypto only needs longer keys.

13 min

2

Post-quantum families

Lattice-based schemes (Kyber, Dilithium), then hash-based, code-based and multivariate.

10 min

3

NIST standardization

Which algorithms are standards today, and the timeline ahead.

5 min

4

Migration challenges

Crypto-agility, certificates and embedded devices.

5 min

5

Exercise

Compare the security assumptions of RSA and a post-quantum algorithm.

7 min

1 The Shor threat model

Data recorded today can be decrypted once a large enough quantum computer exists.

The threat model

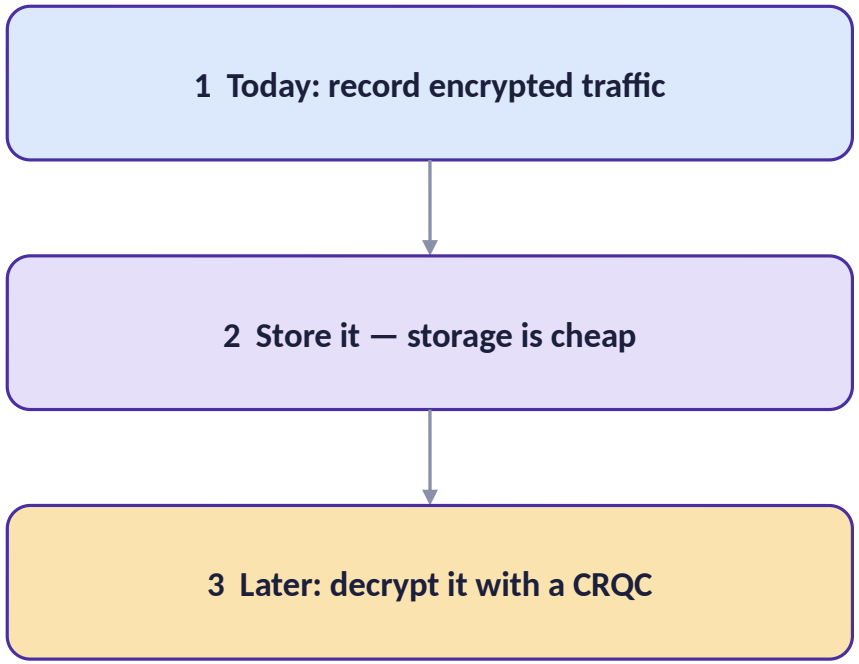
Shor's algorithm, run on a cryptographically relevant quantum computer (CRQC), solves factoring and discrete logarithms in polynomial time.

$$X + Y > Z$$

Mosca's inequality: X = years the data must stay secret, Y = years to migrate, Z = years until a CRQC exists. If X + Y > Z, data encrypted today is already at risk.

Key idea: the deadline is set by how long your data must stay secret, not by when quantum computers arrive.

Harvest now, decrypt later



Signatures are less urgent: forging needs a quantum computer at signing time. Only long-lived ones, such as root certificates and firmware, share the clock.

1 Which classical primitives break

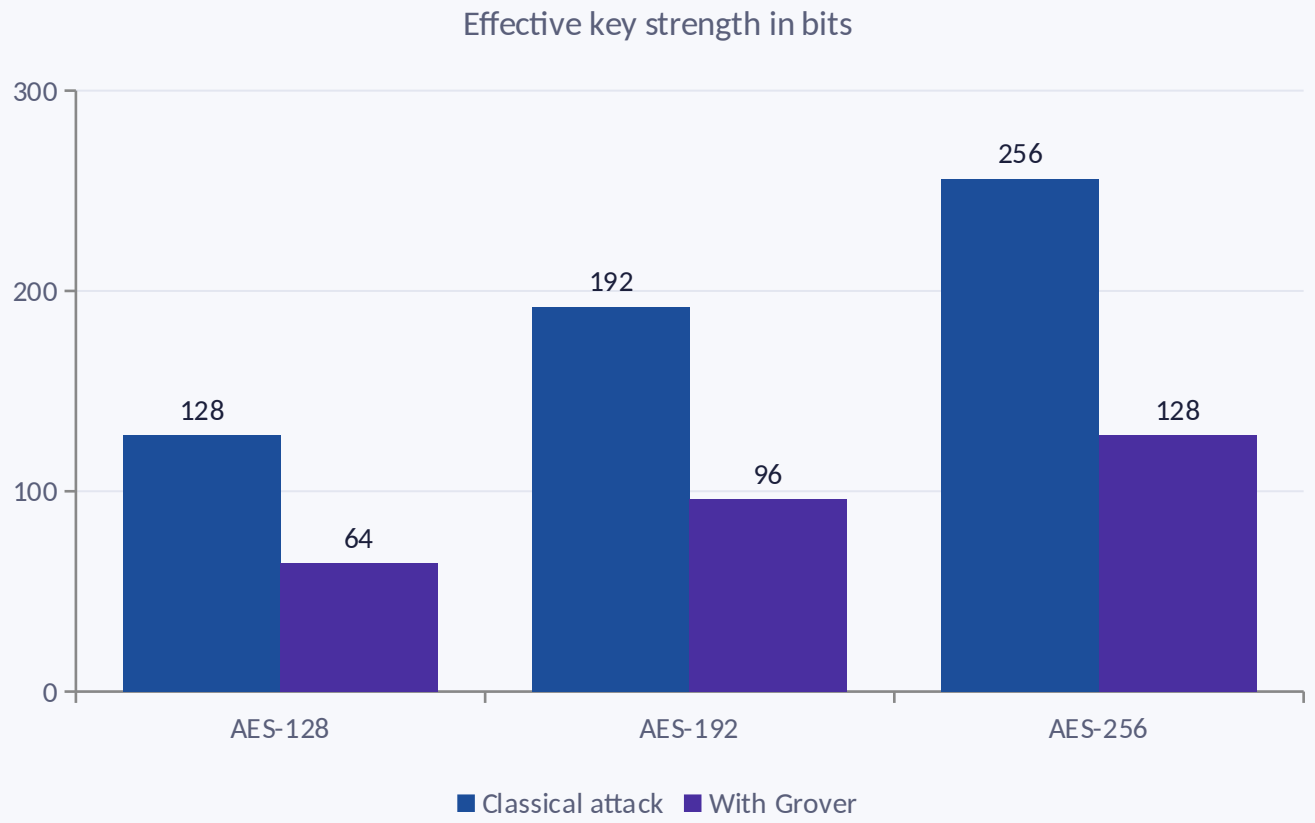
Every widely deployed public-key scheme rests on factoring or discrete logarithms.

Primitive	Hard problem it relies on	Effect of Shor	Replace with
RSA	Integer factoring	**Broken: polynomial time**	ML-KEM, ML-DSA
Diffie–Hellman	Discrete log in a finite field	**Broken: polynomial time**	ML-KEM
ECC (ECDH, ECDSA)	Elliptic-curve discrete log	**Broken: needs fewer qubits than RSA-2048**	ML-KEM, ML-DSA
AES, SHA-2, SHA-3	No algebraic structure for Shor to exploit	**Not broken: only Grover applies**	Keep; use longer keys

Same target, different fix: key exchange and encryption move to a KEM (ML-KEM); signatures and certificates move to ML-DSA or SLH-DSA.

1 Symmetric crypto and hashes just need longer keys

Grover's quadratic speed-up halves the effective key length; it does not break the cipher.



G

Grover halves the exponent

Searching 2^n keys takes about 2^(n/2) quantum steps — a square root, not an exponential break.

A

Prefer AES-256

AES-256 keeps about 128 bits of security. Grover also parallelizes poorly, so real attacks cost more.

H

Hashes: longer outputs

Preimage search also gets a square root. SHA-256 keeps about 2^{128}, and longer outputs add margin.

2 Lattice-based cryptography

Kyber and Dilithium rest on noisy linear equations that no known quantum algorithm solves efficiently.

Learning With Errors (LWE)

Given noisy linear equations mod q, recover the secret vector s. Without the noise it is simple algebra; with small random errors e it appears intractable.

$$b = A \cdot s + e \pmod{q}$$

Module-LWE adds algebraic structure for speed, giving keys of about 1 KB instead of megabytes.

Key idea: the best known attacks, lattice reduction, take exponential time on classical and quantum computers alike.

K

ML-KEM (Kyber)

Key establishment, FIPS 203. At ML-KEM-768: public key 1,184 bytes, ciphertext 1,088 bytes.

D

ML-DSA (Dilithium)

Digital signatures, FIPS 204. At ML-DSA-65: public key 1,952 bytes, signature 3,309 bytes.

↑

The price: bigger objects

X25519 keys are 32 bytes and ECDSA signatures 64 bytes, so every handshake and certificate grows.

2 Hash-based, code-based and multivariate schemes

Three other families rely on very different hard problems, which makes them useful hedges.

H Hash-based

SLH-DSA (FIPS 205, from SPHINCS+); stateful LMS and XMSS (SP 800-208)

Rests on: only the security of a hash function, the most conservative assumption

Trade-off: signatures of about 8–50 KB; stateful schemes need careful key-state handling

Used for: signatures only

C Code-based

Classic McEliece (1978); HQC, selected by NIST in March 2025

Rests on: the difficulty of decoding a random linear code

Trade-off: McEliece has resisted attack for over 40 years, but its public keys are about 260 KB or more

Used for: key exchange (KEMs)

M Multivariate

UOV and relatives, candidates in NIST's additional-signature round

Rests on: solving systems of multivariate quadratic equations

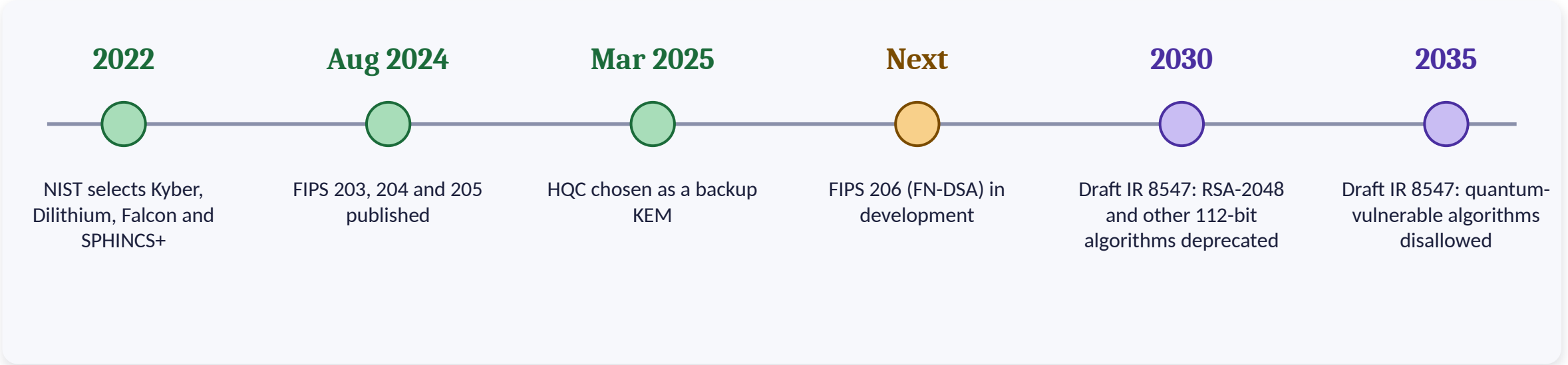
Trade-off: very short signatures, but large public keys and a record of broken schemes

Used for: signatures only

Why diversify? SIKE (isogeny) and Rainbow (multivariate) were both broken on classical computers in 2022; a second family hedges against a surprise in lattices.

3 NIST PQC standardization: status and timeline

Three standards are final; the deprecation dates are still a draft.



FINAL · AUG 2024

ML-KEM, ML-DSA, SLH-DSA

FIPS 203, 204 and 205, derived from Kyber, Dilithium and SPHINCS+: the first post-quantum standards.

IN DEVELOPMENT

FN-DSA (Falcon)

FIPS 206 offers compact lattice signatures, but Falcon is harder to implement safely than ML-DSA.

SELECTED · MAR 2025

HQC, a backup KEM

Code-based, so its security does not depend on lattices. Its standard is still to come.

Status as of October 2026 per NIST CSRC; the 2030 and 2035 dates come from the draft NIST IR 8547.

4 Migration challenges

Swapping algorithms is the easy part; finding, fitting and updating them is hard.

A Crypto-agility

Algorithms hard-coded across systems turn every swap into a rewrite. Negotiate them through clean interfaces so the next change is configuration.

C Certificates

Signatures grow from about 64 bytes (ECDSA P-256) to roughly 2.4–3.3 KB (ML-DSA), so chains get larger and TLS handshakes slow down.

E Embedded devices

Little RAM and flash, and lifetimes of 10–20 years. Firmware signing cannot wait: stateful LMS and XMSS are already approved for it.

A practical path

1 Inventory algorithms and keys → 2 Protect long-lived data first → 3 Deploy hybrid (classical + PQ) → 4 Retire classical algorithms

5 Exercise: compare the security assumptions

Work in pairs for 5 minutes: fill in the ML-KEM-768 column, then we compare answers.

Task: RSA-2048 and ML-KEM-768 can both establish a shared secret. For each question, say what each one relies on and how a quantum attacker changes the picture.

Question	RSA-2048	ML-KEM-768 (your answer)
Hard problem	Factoring N = p·q	*Your answer…*
Best classical attack	Number field sieve (sub-exponential)	*Your answer…*
Best known quantum attack	Shor: polynomial time	*Your answer…*
How long studied	Since the 1970s	*Your answer…*
Public key size	256 bytes	*Your answer…*

Bonus: which assumption would you trust for data that must stay secret for 30 years, and why?

5 Exercise answers: two different bets

RSA bets on one problem that Shor solves; ML-KEM bets on a younger problem with no known quantum shortcut.

Question	RSA-2048	ML-KEM-768
Hard problem	Factoring N = p·q (the RSA problem)	Module-LWE: recover s from (A, A·s + e mod q)
Best classical attack	Number field sieve: sub-exponential time	Lattice reduction (BKZ, sieving): exponential time
Best known quantum attack	Shor: polynomial time, so broken	No efficient algorithm known; sieving gains are modest
How long studied	Since the 1970s (RSA, 1977)	Since the 1990s (Ajtai 1996, Regev's LWE 2005)
Public key size	256 bytes	1,184 bytes

RSA

One problem, and a quantum algorithm that solves it outright.

ML-KEM

No known quantum attack, but a shorter track record of analysis.

Both

Neither assumption is proven, which is why hybrids combine them.

ONE-LINE TAKEAWAY

Shor breaks RSA, Diffie–Hellman and ECC, while Grover only asks for longer symmetric keys; lattice-based ML-KEM and ML-DSA are the new defaults, so start migrating now.

Shor: public-key breaks

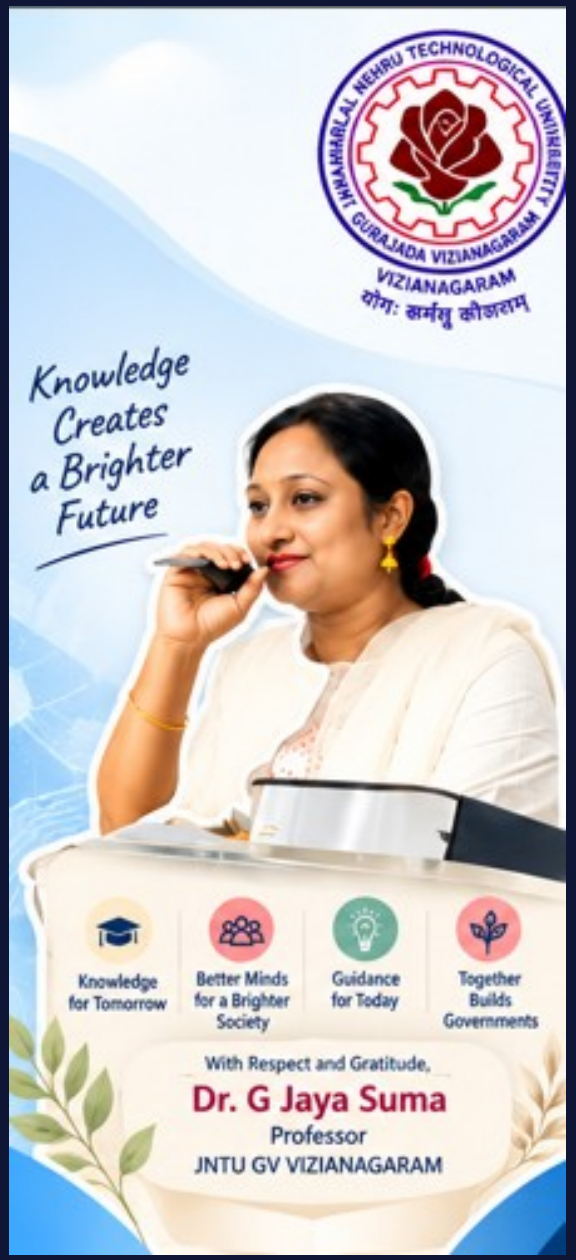
Grover: longer keys

Migrate: crypto-agility

→

Next: Quantum communication and key distribution

We look at the other defence against quantum attackers: using quantum physics itself to distribute keys.



JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY
GURAJADA VIZIANAGARAM

TECHNOLOGY FOR A BETTER TOMORROW

Learn
Innovate
Create
Lead

VIDEO LECTURE SERIES · QUANTUM ALGORITHMS

Quantum Machine Learning

Parameterized circuits, quantum kernels and the hybrid training loop: where they might help, and what the evidence shows so far.

55 MIN

Encoding | Variational | Kernels | Evidence

Fundamentals of Quantum Computing · Quantum Algorithms: Quantum Machine Learning

A

What we'll cover in 55 minutes

Five segments, from where quantum might help to what the benchmarks actually show.

1

Where quantum could help

Data encoding, kernels and optimization, plus basis, amplitude and angle encoding.

10 min

2

Variational circuits and VQE

Parameterized circuits as models, the hybrid training loop, and the eigensolver.

14 min

3

Quantum kernel methods

Feature maps, kernel estimation and quantum support vector machines.

8 min

4

A toy classifier, step by step

Build and train a variational classifier on a small dataset.

10 min

5

Obstacles and evidence

Barren plateaus, other training problems, and quantum vs classical benchmarks.

13 min

1 Where quantum could help ML

Three places a quantum device might add something a classical model cannot easily do.

In simple words

A quantum model maps data into the state space of qubits, processes it with gates, and reads out measurement statistics. The hoped-for gain comes from the three places on the right.

data → encode → circuit → measure

Caveat: loading classical data and reading out results are costly, so a speed-up inside the circuit can vanish end to end.

E

Data encoding

A feature map places data in an exponentially large Hilbert space where patterns may be easier to separate.

K

Kernels

Overlaps between encoded states define kernels that might be hard to compute classically.

O

Optimization

A parameterized circuit is trained like a neural network, with a quantum device evaluating the cost.

1 Classical data encoding

Three standard ways to turn numbers into qubit states, each with a different cost.

B Basis encoding

$$5 \rightarrow |101\rangle$$

Qubits: one qubit per bit of each value

Circuit cost: shallow: just X gates

Best for: binary data; one sample at a time

A Amplitude encoding

$$\mathbf{x} \rightarrow \Sigma\, x_i\, |i\rangle \;/\; \|\mathbf{x}\|$$

Qubits: $\log_2 N$ qubits for N features

Circuit cost: state preparation is deep in general

Best for: compact storage, if loading is cheap

θ Angle encoding

$$\mathbf{x} \rightarrow \mathrm{RY}(\mathbf{x})|0\rangle$$

Qubits: one qubit per feature

Circuit cost: shallow and hardware-friendly

Best for: near-term variational models

Trade-off: amplitude encoding is compact in qubits but expensive to prepare; angle encoding is cheap but needs a qubit per feature.

2 Parameterized circuits as models

A circuit with tunable rotation angles defines a function of the data that we can train.

The model

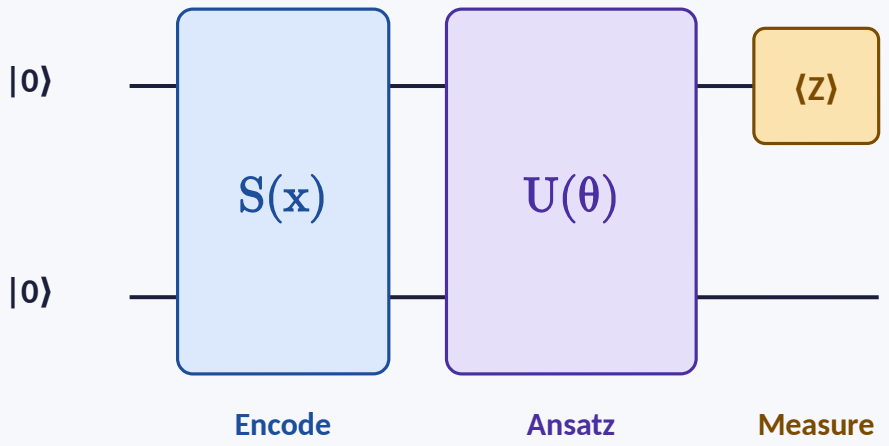
$$f(x;\,\theta) = \langle\psi|\, O \,|\psi\rangle$$

Here $|\psi\rangle = U(\theta)\, S(x)\, |0\ldots 0\rangle$: encode x with S(x), apply a trainable ansatz U(θ), then measure an observable O such as Z on one qubit.

Not: a fixed algorithm with a known answer. **Instead:** a trainable function, like a small neural network.

Key idea: the ansatz must be expressive enough to fit the data, yet shallow enough to run and train on real hardware.

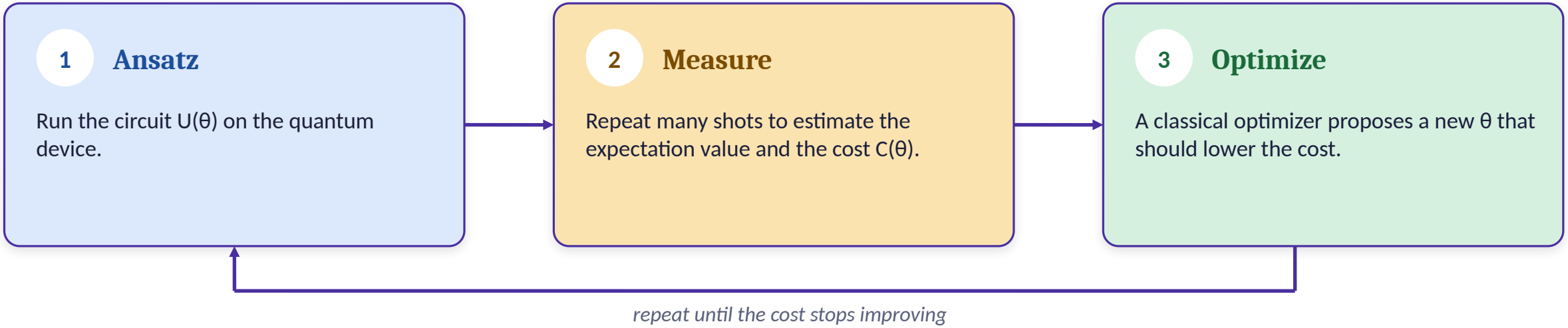
A two-qubit model



The prediction is the measured expectation value; the angles θ are what training adjusts.

2 The variational algorithm loop

A quantum device evaluates the cost; a classical optimizer decides what to try next.



H Hybrid by design

Only the cost evaluation runs on quantum hardware; everything else stays classical.

S Shallow circuits

Short circuits suit noisy near-term devices, which is why this recipe dominates today.

∂ Gradients from the device

Parameter-shift: $\partial f/\partial\theta = \tfrac{1}{2}\,[f(\theta+\pi/2) - f(\theta-\pi/2)]$, two extra circuit runs per parameter.

2 Variational Quantum Eigensolver (VQE)

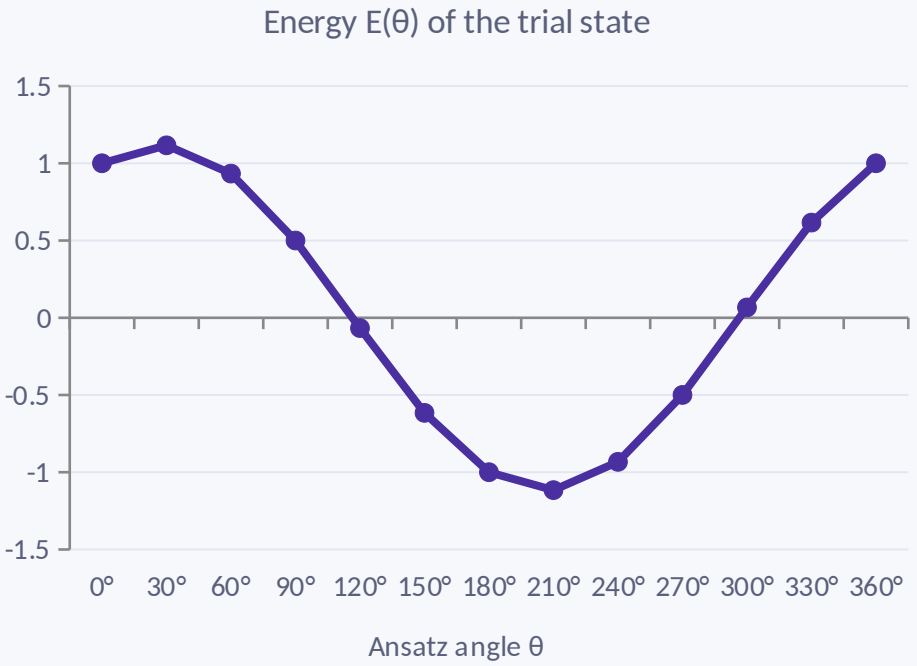
Minimize the energy of a trial state to approach the ground state of a Hamiltonian.

The variational principle

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle \geq E_0$$

Any trial state has energy at or above the true ground-state energy E_0, so lowering $E(\theta)$ approaches E_0 from above. H is split into Pauli terms, each measured separately.

Where it is used: molecular ground states in chemistry and materials. The same loop, with a different cost, trains the ML models in this lecture.



Toy case: $H = Z + 0.5\,X$ with the ansatz $RY(\theta)|0\rangle$. The lowest point is the exact ground energy, $-\sqrt{1.25} \approx -1.118$.

3 Quantum kernels and quantum SVMs

The quantum computer estimates similarities between data points; a classical SVM does the rest.

The quantum kernel

$$k(x, x') = |\langle \varphi(x) | \varphi(x') \rangle|^2$$

Run $U(x')^\dagger U(x)$ on $|0\ldots 0\rangle$ and count how often the result is all zeros: that frequency is $k(x, x')$. Fill a table of these values and give it to an ordinary SVM.

Catch: an advantage needs a feature map that is hard to simulate classically yet still fits the data. Proven speed-ups exist only for contrived datasets (Liu et al., 2021).

The QSVM recipe

1 Encode x and x′ with a feature map

2 Estimate the overlap by measuring

3 Fill the Gram matrix for all pairs

4 Train a classical SVM on it

N training points need about $N^2/2$ kernel estimates, each from many shots, so cost grows quickly with dataset size.

4

Building a variational classifier on a toy dataset

Two-moons data, two qubits and six trainable parameters, simulated exactly.

1

Data

Two-moons, 200 points scaled to [0, π]: 140 train, 60 test.

2

Encode

Angle encoding: RY(x_1) and RY(x_2) on two qubits.

3

Ansatz

Three layers of RY rotations plus a CNOT: 6 parameters.

4

Train

Measure ⟨Z⟩ on qubit 0, minimise squared loss with COBYLA, best of 5 starts.



90%

Variational circuit

90%

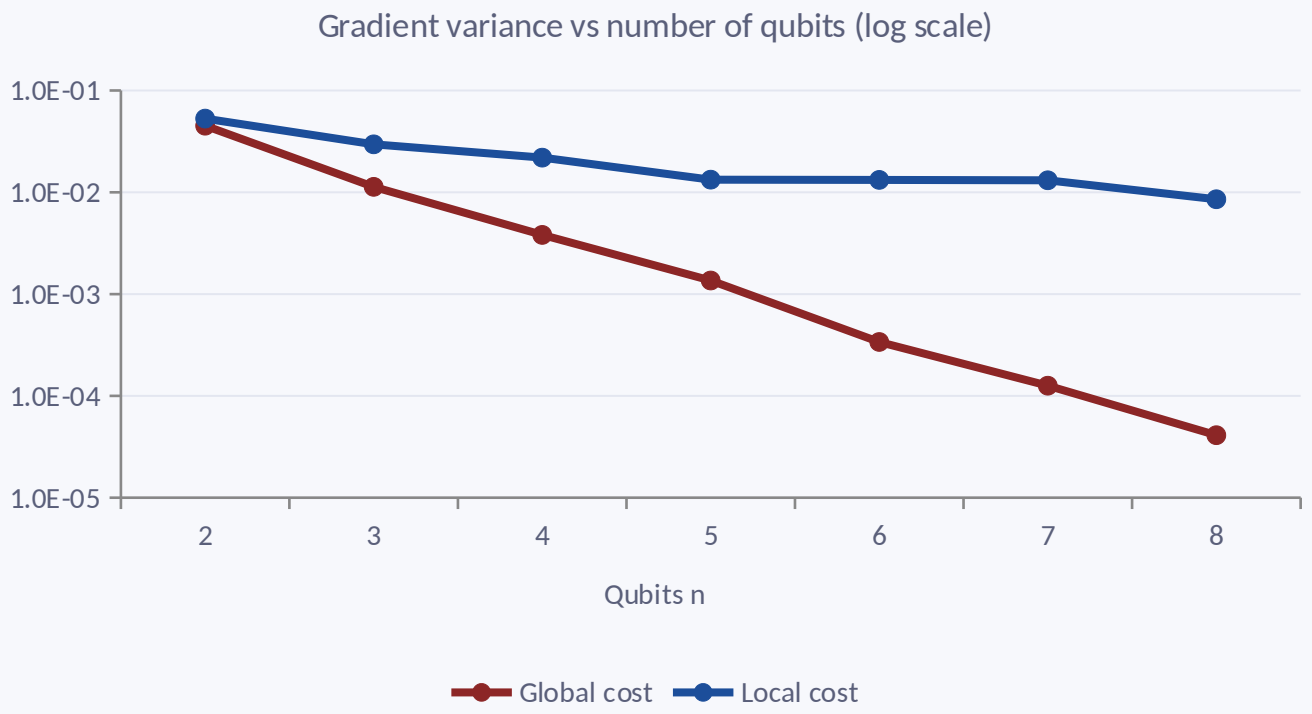
Logistic regression

100%

RBF SVM

5 Barren plateaus and other training obstacles

On a barren plateau the cost landscape is almost flat, so gradients drown in measurement noise.



Our simulation: random RY + CZ circuits with n layers, 300 samples per point. The global cost shrinks about 3× per added qubit.

≈

Flat landscapes

For deep random circuits and global costs, gradients vanish exponentially with qubit count (McClean et al., 2018).

N

Noise and shots

Hardware noise can flatten the landscape too, and each gradient needs many measurement shots.

\+

Mitigations

Use local cost functions, shallow structured ansatzes, smart initialization or layer-by-layer training.

5 Quantum vs classical on small datasets

The largest systematic benchmark so far found no advantage for quantum models.

12

quantum ML models

6

binary classification tasks

160

datasets in total

Bowles, Ahmed and Schuld (2024), "Better than classical?"

1 What they found

Out-of-the-box classical models generally beat the quantum classifiers, and removing entanglement often gave equal or better results.

2 Why be cautious

Some proposed speed-ups were later matched by "dequantized" classical algorithms (Tang, 2019), and small test sets are noisy.

3 What would change it

Tuned classical baselines, honest hyperparameter search, and tasks with quantum structure, such as data from quantum experiments.

ONE-LINE TAKEAWAY

Quantum ML trains parameterized circuits in a hybrid loop, but training is hard and small-data benchmarks show no reliable advantage over tuned classical models yet.

Encode: basis, amplitude, angle

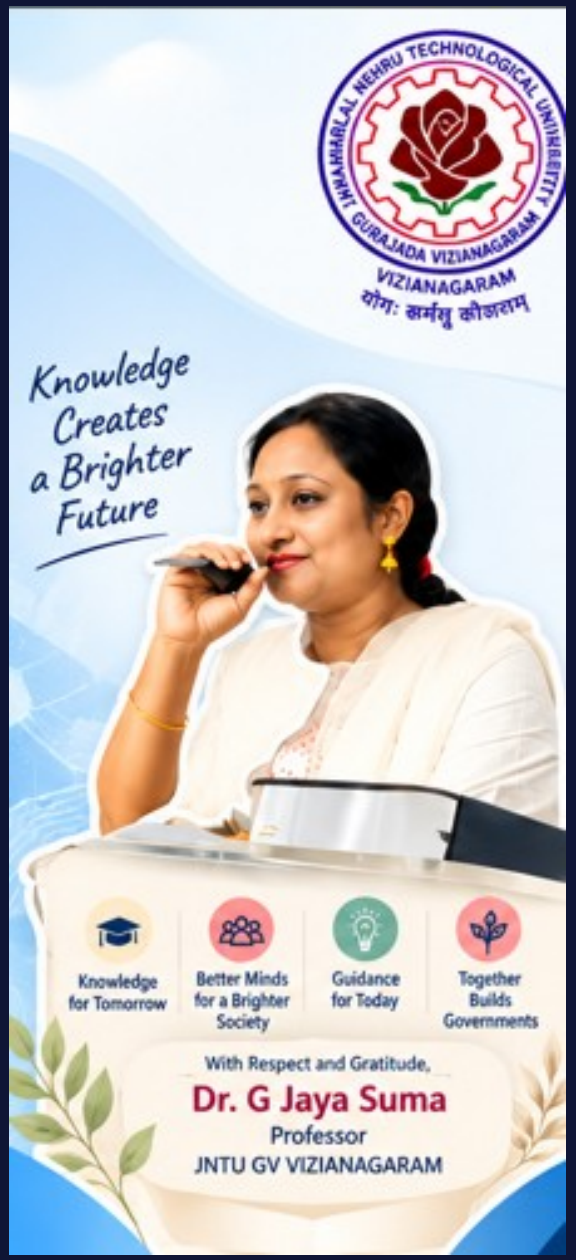
Train: ansatz, measure, optimize

Evidence: no proven edge yet

→

Next: Noise, error mitigation and the road to fault tolerance

We look at why noise limits today's circuits and what it takes to run deeper algorithms reliably.



JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY
GURAJADA VIZIANAGARAM

TECHNOLOGY FOR A BETTER TOMORROW

Learn
Innovate
Create
Lead

VIDEO LECTURE SERIES · QUANTUM ALGORITHMS

Optimization and Quantum Annealing

From QUBO and Ising models to annealers and QAOA: how optimization problems are posed for quantum hardware, and when that is worth doing.

45 MIN

QUBO | Annealing | QAOA | Applications

Fundamentals of Quantum Computing · Quantum Algorithms: Optimization and Quantum Annealing

A

What we'll cover in 45 minutes

Five segments, from posing a problem as an energy to judging whether quantum hardware suits it.

1

Formulating optimization problems

Combinatorial problems and their QUBO and Ising forms.

7 min

2

Annealing and D-Wave

The adiabatic principle, then D-Wave architecture, embedding and limits.

10 min

3

QAOA and a Max-Cut example

Cost and mixer layers, with a worked five-node graph.

12 min

4

Finance and logistics

Portfolio optimization, risk analysis, routing and scheduling.

8 min

5

Exercise

Pick a problem suited to annealing and justify the choice.

8 min

1 Combinatorial optimization: QUBO and Ising

Write the cost of every choice as an energy over binary variables, then look for the lowest energy.

Two equivalent forms

Combinatorial optimization means picking the best option from a huge discrete set: routes, schedules, portfolios. Quantum hardware takes the cost as an energy.

QUBO (binary variables)

$$\min \; x^{T} Q\, x\,, \quad x_i \in \{0, 1\}$$

Ising model (spins)

$$H = \Sigma\, J_{ij}\, s_i\, s_j + \Sigma\, h_i\, s_i\,, \quad s_i \in \{-1, +1\}$$

Substituting $x = (1 - s)/2$ converts one form into the other, so a problem written either way runs on both annealers and QAOA.

M

Max-Cut as an example

Split a graph's nodes into two sets to cut as many edges as possible: minimise $\Sigma\, s_i\, s_j$ over its edges.

P

Constraints become penalties

A rule such as "pick exactly K items" turns into a term $P(\Sigma x_i - K)^2$ that adds energy when it is broken.

H

Hard in general

These problems are NP-hard, so quantum hardware offers heuristics that search for low energy, not guaranteed optima.

2 Adiabatic computing and the annealing principle

Start in an easy ground state and change the Hamiltonian slowly toward the problem you want to solve.

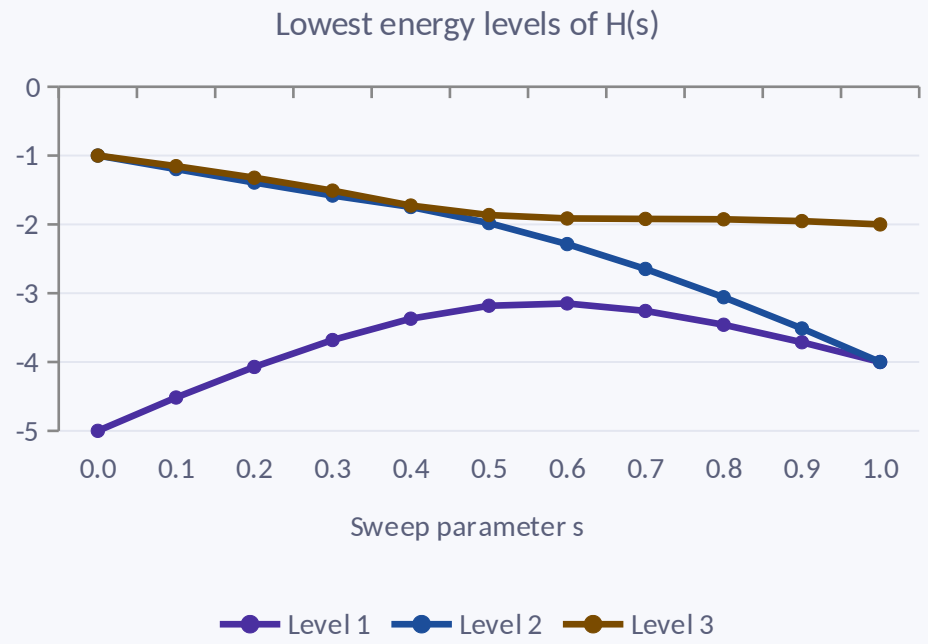
The interpolation

$$H(s) = (1 - s)\, H_D + s\, H_P$$

H_D is a driver with an easy ground state (all qubits in $|+\rangle$); H_P encodes the problem so that its ground state is the best answer. Sweep s from 0 to 1.

Not: testing candidate answers one by one. **Instead:** letting the physics settle into low energy.

Adiabatic theorem: if the sweep is slow compared with the inverse square of the smallest energy gap, the system stays in the ground state. Real annealers are noisy and finite-speed.



Computed for the 5-node Max-Cut instance used later. At s = 1 the two lowest levels meet: both are optimal cuts. Level 3 stays at least 1.2 above the ground level.

2 D-Wave hardware: architecture, embedding, limits

A superconducting annealer with thousands of qubits, but sparse connections between them.

4,400+

qubits in Advantage2

20

connections per qubit (Zephyr)

May 2025

Advantage2 generally available

A Architecture

Superconducting loops act as qubits, and the annealing schedule is controlled by the user. It solves QUBO and Ising problems only, with no general gates.

E Embedding

Your problem graph rarely matches the hardware graph, so each variable becomes a chain of physical qubits tied together. Long chains use up qubits and add noise.

L Limitations

The hardware is analog, noisy and limited in precision. Speed-ups over the best classical solvers have been claimed only for specific benchmarks and remain debated.

3 QAOA: cost and mixer layers

A gate-based cousin of annealing: alternate two operators, then tune their angles classically.

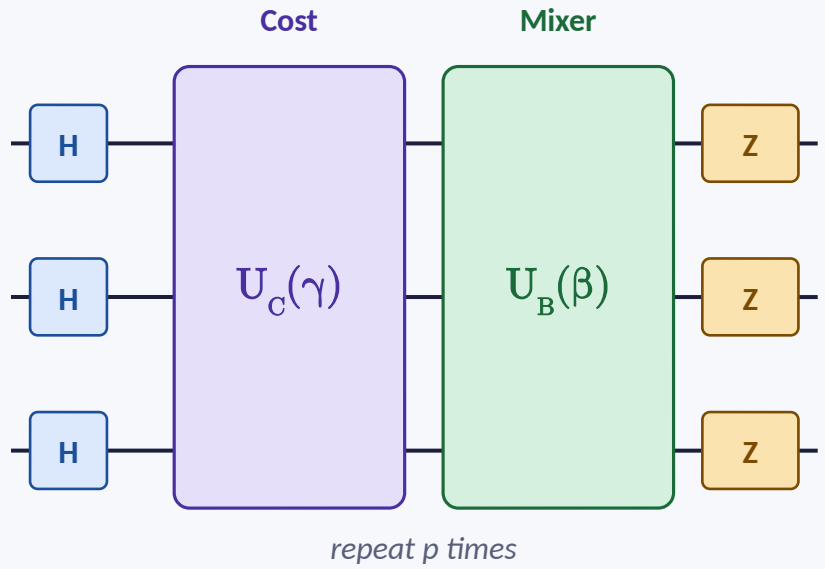
The ansatz

$$|\gamma, \beta\rangle = \Pi\, e^{-i\beta B}\, e^{-i\gamma C}\, |+\rangle^{n}$$

Start in the uniform superposition $|+\rangle^{n}$. Each of p layers applies a cost operator, then a mixer. Measure, evaluate the cost, and let a classical optimizer adjust γ and β.

Link to annealing: QAOA is a discretized annealing schedule with p steps. As p grows it can follow the adiabatic path, but circuits get deeper.

One QAOA layer on three qubits



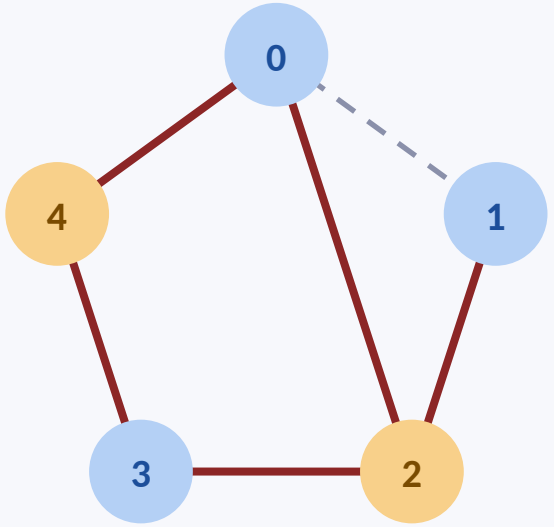
Cost layer: a ZZ phase rotation on every edge (i, j)

Mixer layer: an X rotation on every qubit

3 Max-Cut worked example

Split five nodes into two sets to cut as many of the six edges as possible.

The graph and its best cut



Blue {0, 1, 3} vs amber {2, 4}. Red edges are cut (5 of 6); the dashed edge is not.

$$\text{cut}(s) = \Sigma\ (1 - s_i\, s_j)\ /\ 2 \quad \text{over edges}$$

Random guess

60%

of the best cut, on average

Average over all 32 assignments

QAOA, p = 1

82%

of the best cut

41% chance of an optimal cut

QAOA, p = 2

92%

of the best cut

71% chance of an optimal cut

QAOA, p = 3

97%

of the best cut

85% chance of an optimal cut

4 Portfolio optimization and risk analysis

Choosing assets is a QUBO; estimating risk is a different, longer-term quantum opportunity.

Mean–variance selection

$$\min \; q\, x^{T} \Sigma\, x - \mu^{T} x \; + P(\Sigma x_i - K)^2$$

Let $x_i = 1$ if asset i is held. The covariance Σ supplies the pairwise couplings, μ rewards return, and the budget rule “hold exactly K assets” becomes a penalty.

Honest status: plain continuous mean–variance problems are solved classically at large scale. The quantum case rests on hard variants with cardinality limits, lot sizes and trading costs.

x

Binary variables

Pick or skip each asset. Finer weights need several bits per asset, so qubit counts grow quickly.

Σ

Risk as couplings

Every covariance between two assets is a coupling between two qubits, so the problem is dense and embedding is costly.

V

Risk analysis

Amplitude estimation could give a quadratic speed-up for Monte Carlo measures such as value at risk, but needs fault-tolerant hardware.

4 Logistics, routing and supply-chain scheduling

Natural problems, but the binary encoding grows fast and constraints dominate.

R Routing

Vehicle routing and travelling-salesman problems: which stop follows which, under capacity and time limits.

S Scheduling

Job-shop and shift scheduling: assign tasks to machines and time slots without clashes.

C Supply chain

Facility location and inventory decisions: which sites to open and how to serve demand at least cost.

Binary variables for a travelling-salesman tour (one per city and position, n^2)

25	100	400	1,600
5 cities	10 cities	20 cities	40 cities

Reality check

One-hot rules need penalty terms that shape a rugged landscape. Today these problems usually run on hybrid solvers, against strong classical baselines such as OR-Tools or Gurobi.

5 Exercise: is this problem suited to annealing?

Work in pairs for 5 minutes: judge each candidate with the checklist and justify your answer.

Suitability checklist

1. Decisions are binary, or need only a few bits
2. The objective is quadratic (pairwise interactions)
3. Constraints can be written as penalty terms
4. The size fits the hardware after embedding
5. A classical baseline exists to compare against

A Select 10 of 40 stocks to minimise portfolio variance

B Solve a linear program with 10,000 continuous variables

C Schedule 8 jobs on 3 machines with deadlines

Task: mark each candidate good fit, poor fit or maybe, and cite the checklist items behind your verdict.

5 Exercise answers: justify the verdict

A good answer names the checklist items and a classical baseline, not just a yes or no.

GOOD FIT

A · Stock selection

Binary pick or skip variables, a quadratic risk term from the covariance matrix, and one budget rule that becomes a single penalty. 40 variables is small enough to embed on current hardware.

POOR FIT

B · Linear program

The variables are continuous and the objective is linear. Simplex and interior-point methods already solve this quickly, and discretizing would waste qubits.

MAYBE

C · Job scheduling

It can be encoded with one-hot job, machine and time-slot variables, but deadlines and clashes need many penalty terms. Compare with a constraint solver such as OR-Tools CP-SAT first.

Rule of thumb: binary, quadratic, constraint-light and modest in size, then test it against the best classical solver before claiming any benefit.

ONE-LINE TAKEAWAY

Write the problem as a QUBO or Ising energy, then let an annealer or QAOA search for low energy; choose problems that are binary, quadratic and small, and always compare with classical solvers.

Formulate: QUBO and Ising

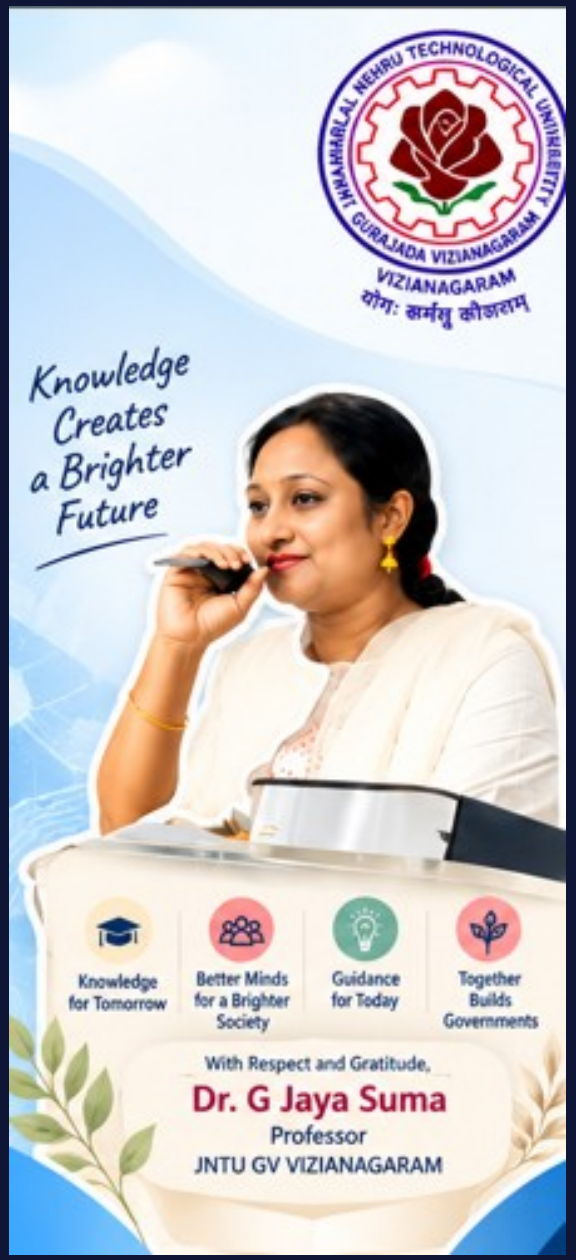
Search: anneal or QAOA

Check: classical baseline

→

Next: Quantum simulation and chemistry

We turn to the problem quantum computers were first proposed for: simulating quantum systems themselves.



JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY
GURAJADA VIZIANAGARAM

TECHNOLOGY FOR A BETTER TOMORROW

Learn
Innovate
Create
Lead

VIDEO LECTURE SERIES · QUANTUM APPLICATIONS

Quantum Chemistry and Materials Simulation

Why simulating nature was the first dream of quantum computing, and how far it has really come.

40 MIN

Fermions → qubits | VQE | Drug design | Materials | Reality check

0 What we'll cover in 40 minutes

Seven segments, from Feynman's idea to a realistic view of today's results.

1	**Feynman's motivation** Why simulating quantum systems needs quantum machines.	5 min
2	**The electronic-structure problem** What we must solve, and why classical cost explodes.	6 min
3	**Fermions to qubits** Jordan–Wigner and Bravyi–Kitaev transforms.	8 min
4	**Walkthrough: H_2 with VQE** Find the ground-state energy of hydrogen step by step.	8 min
5	**Drug discovery pipelines** Where accurate molecular energies could fit.	4 min
6	**Batteries, catalysts and materials** Use cases and why they are hard.	4 min
7	**Realistic assessment** Demonstrated today versus projected for later.	5 min

1

Feynman’s motivation: simulate quantum with quantum

Nature is quantum, and a classical computer pays an exponential price to imitate it.

The cost of brute force

n spin-½ particles need 2^n complex amplitudes. At 16 bytes each, the memory needed is:

17 GB	18 TB	18 PB
30 particles	40 particles	50 particles

Every extra particle doubles the cost, so exact simulation stops at roughly 50 particles on the largest machines.

Feynman’s answer: if nature is quantum, build the simulator out of quantum parts.

1981

The proposal

In “Simulating Physics with Computers”, Feynman argued that classical machines cannot efficiently imitate quantum behaviour, and proposed quantum computers.

1996

It can work

Seth Lloyd showed that a quantum computer can simulate any system with local interactions efficiently, with cost growing polynomially.

Now

Chemistry and materials first

The electrons in molecules and solids are exactly the quantum system, so they are the most natural target.

2 The electronic-structure problem

Find the lowest energy of the electrons for fixed nuclei; exact methods scale badly.

The electronic Hamiltonian (second quantised)

$$H = \sum h_{pq}\, a^\dagger_p a_q + \tfrac{1}{2} \sum h_{pqrs}\, a^\dagger_p a^\dagger_q a_r a_s$$

The first term is electrons moving in the field of the nuclei. The second is electron–electron repulsion. With N spin orbitals there are about N^4 integrals h.

Target: chemical accuracy, about 1.6 mHa (1 kcal/mol). Reaction rates and binding strengths depend on energy differences that small.

Classical methods trade cost for accuracy

Method	Typical cost	The catch
Hartree–Fock	$\sim N^4$	Ignores electron correlation
DFT	$\sim N^3$–N^4	Approximate functional; unreliable when correlation is strong
CCSD(T)	$\sim N^7$	Gold standard, but fails for strongly correlated systems
Full CI (exact)	exponential	Only for very small systems

Cost grows with N, the number of basis functions. Quantum algorithms aim to reach full-CI accuracy at polynomial cost.

3

Mapping fermions to qubits: Jordan–Wigner

Electrons are fermions, so a Z string restores the signs that plain qubits lack.

Why a mapping is needed

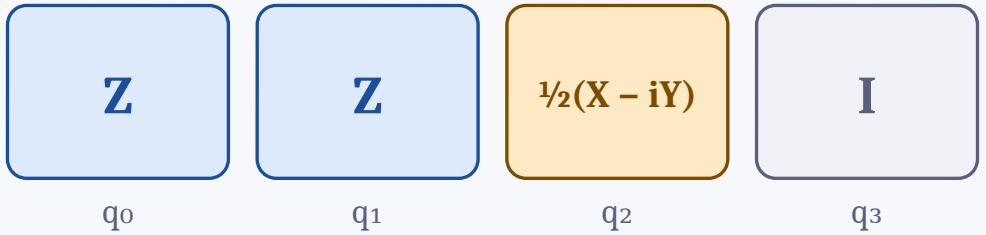
Swapping two fermions flips a sign: $a^\dagger_p a^\dagger_q = -a^\dagger_q a^\dagger_p$. Operators on different qubits simply commute, so the signs must be added by hand.

$$a^\dagger_j = Z_0 \otimes \ldots \otimes Z_{j-1} \otimes \tfrac{1}{2}(X_j - iY_j)$$

qubit j = $|1\rangle$ means spin orbital j is occupied

Simple and direct: each qubit stores one occupation, and the number operator is just $n_j = \tfrac{1}{2}(I - Z_j)$.

Example: $a^\dagger_2$ on four qubits



The Z string on qubits below j counts how many electrons the operator passes, giving the right sign.

Good

Occupation is local to one qubit.

Costly

String length grows with j: up to n qubits per term.

3 Bravyi–Kitaev: shorter Pauli strings

Store partial parities in a tree so each operator touches only about log n qubits.

Idea: store parities, not occupations

Each qubit holds the parity of a block of orbitals. Updating an occupation and summing signs then both touch only O(log n) qubits.

q_0	q_1	q_2	q_3
n_0	$n_0 \oplus n_1$	n_2	$n_0 \oplus n_1 \oplus n_2 \oplus n_3$

Four spin orbitals; ⊕ means addition mod 2.

For H_2 (4 spin orbitals) either map uses 4 qubits. Symmetries then shrink the problem to 2 qubits.

Jordan–Wigner vs Bravyi–Kitaev

	Jordan–Wigner	Bravyi–Kitaev
Qubit j stores	Occupation of orbital j	Parity of a block of orbitals
Pauli string length	Up to n	About log n
Reading an occupation	One qubit	About log n qubits
Best for	Small molecules, simplicity	Larger systems, shallower circuits

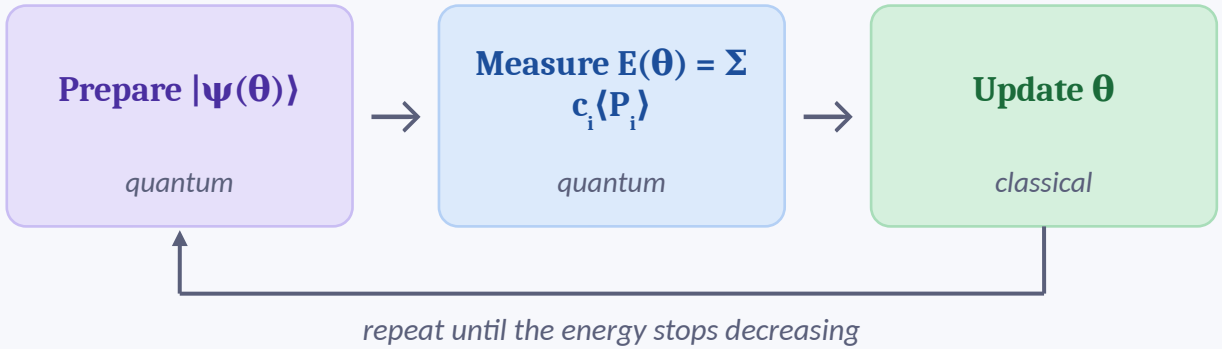
Both give identical energies; they differ only in circuit cost.

4

Walkthrough: ground-state energy of H_2 with VQE

A quantum circuit proposes states and a classical optimiser steers them downhill in energy.

The VQE loop



Variational principle: $E(\theta) \geq E_0$ for every θ, so the lowest energy found is an upper bound on the ground state.

Short circuits plus a classical optimiser make VQE a near-term algorithm. The price is many repeated measurements.

H_2 set-up

Molecule: H_2 at 0.735 Å in the STO-3G basis.

Qubits: 4 spin orbitals give 4 qubits; symmetries cut this to 2.

Ansatz: Start in the Hartree–Fock state, then rotate by one angle θ.

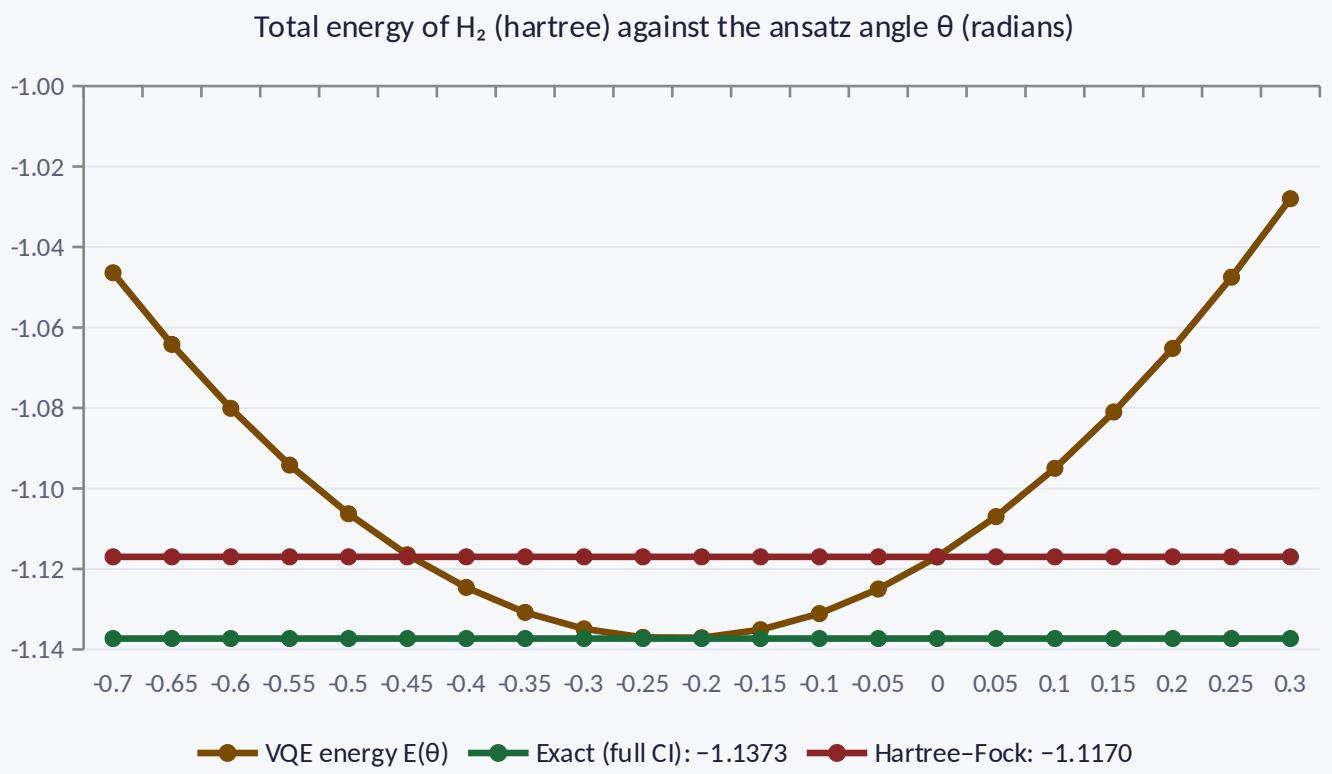
$$H = -1.0524\ II + 0.3979\ IZ - 0.3979\ ZI - 0.0113\ ZZ + 0.1809\ XX$$

Electronic energy in hartree; add 0.7200 for nuclear repulsion.

4

Result: VQE reaches the exact H_2 energy

One parameter is enough here: scanning θ finds the minimum below Hartree–Fock.



Noise-free simulation of the 2-qubit circuit. On real hardware, gate and shot noise shift these values.

−1.1373 Ha

Lowest VQE energy, which equals the exact answer

20 mHa

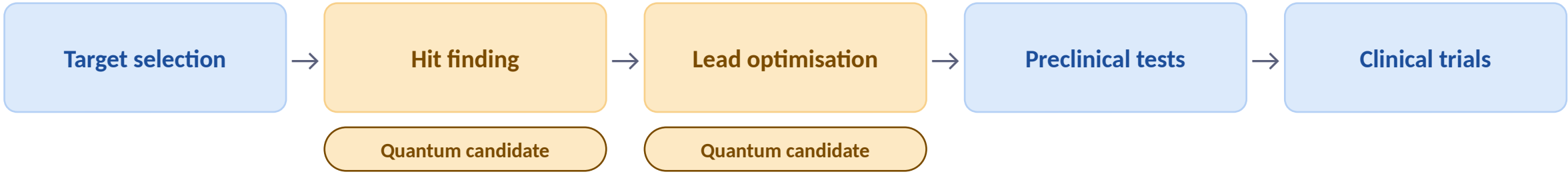
Correlation energy recovered beyond Hartree–Fock

1.6 mHa

Chemical accuracy: the target for any hardware run

5 Drug discovery and molecular design pipelines

Quantum simulation is a candidate accuracy step inside a mostly classical pipeline.



E Binding energies

Whether a drug binds is decided by small energy differences. An error of about 1.4 kcal/mol changes the predicted binding strength about tenfold.

M Hard chemistry

Metal-containing enzymes such as cytochrome P450, which metabolises many drugs, have strongly correlated electrons that standard DFT handles poorly.

P How it fits

Quantum would solve a small active-site model accurately, then feed the result back into classical docking, force fields and machine learning.

Status: drug-related runs on quantum hardware are proofs of concept on small fragments, not yet part of routine drug design.

6 Battery, catalyst and materials research

Each field needs accurate energies for electrons that interact strongly.

B Batteries

Question: Which electrolytes and electrodes store more energy, charge faster and last longer?

Why it is hard: Electrolyte breakdown and lithium–sulfur or lithium–air reactions involve bond breaking that approximate methods struggle with.

C Catalysts

Question: Can we find cheaper catalysts for fertiliser and CO_2 conversion?

Why it is hard: Nitrogenase fixes nitrogen at room temperature using an iron–sulfur–molybdenum cluster, a benchmark strongly correlated problem. Haber–Bosch uses about 1–2% of world energy.

M Materials

Question: Why do some materials superconduct at high temperature, and how do exotic magnetic phases form?

Why it is hard: Lattice models such as the Hubbard model map naturally onto qubits, so quantum simulators already probe them.

Common thread: strongly correlated electrons, where DFT is least reliable and exact classical methods are out of reach.

7 Realistic assessment: demonstrated vs projected

Small, checked results exist today; chemically useful advantage is still ahead.

Demonstrated

Ground-state energies of H_2, LiH and BeH_2 with VQE on small processors.

Water and similar molecules in minimal bases on up to about 16 qubits.

Answers match exact classical results, because the problems are small enough to check.

Emerging

Hybrid workflows: the quantum chip samples configurations, a classical computer diagonalises.

Runs on 100+ qubit devices for fragments of larger molecules, still verified classically.

Error mitigation tames noise but costs many extra measurements.

Projected

Phase-estimation energies for FeMoco- and P450-size active sites.

Needs error-corrected machines; published resource estimates run to millions of physical qubits.

Practical advantage over improving classical methods is still unproven.

Rule of thumb: judge a claim by what was run on hardware and verified, not by the molecule named in the headline. Status as of late 2026; check recent papers.

ONE-LINE TAKEAWAY

Simulating molecules is the most natural job for a quantum computer, but today's results are small proofs of concept and real impact awaits fault-tolerant hardware.

Fermions → qubits: JW, BK

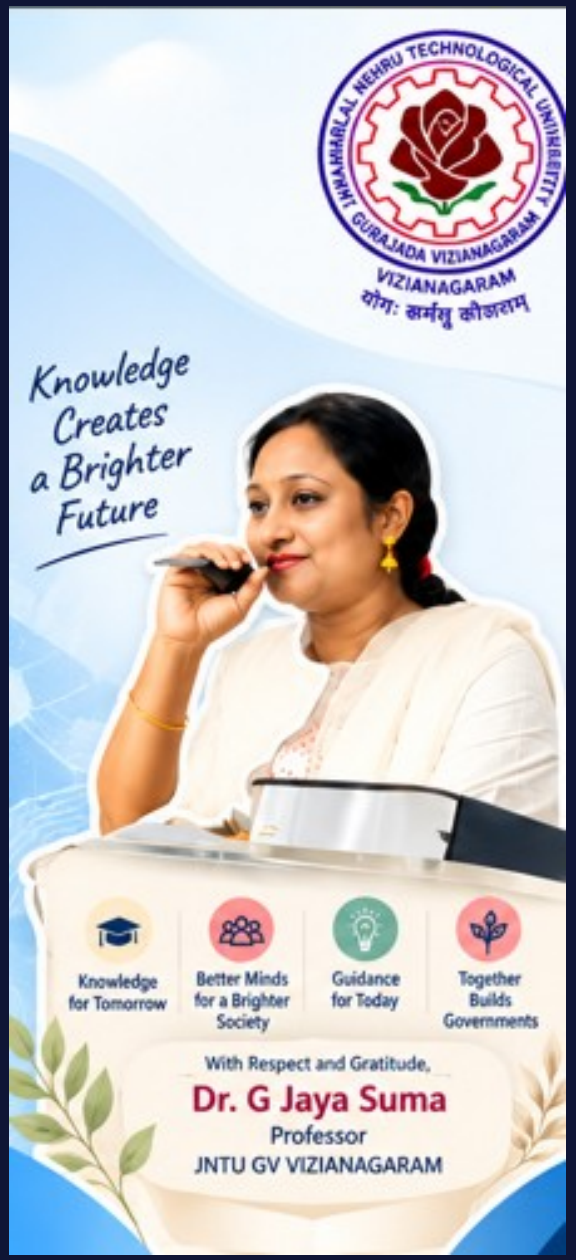
VQE solved H_2 exactly

Advantage still projected

→

Next: Other application areas

We move from simulating nature to other problems where quantum methods may help.



JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY
GURAJADA VIZIANAGARAM

TECHNOLOGY FOR A BETTER TOMORROW

Learn
Innovate
Create
Lead

VIDEO LECTURE SERIES · QUANTUM APPLICATIONS

Industry Landscape and Career Pathways

Who is building quantum computers, what governments are funding, and how you can join the field.

30 MIN

Hardware players | National missions | Roadmaps | Careers | Further study

0 What we'll cover in 30 minutes

Six segments, from the companies building machines to your own next steps.

1	**Hardware players** IBM, Google, IonQ, Quantinuum, Rigetti, PsiQuantum and QuEra.	6 min
2	**National missions and funding** India's National Quantum Mission and programmes elsewhere.	5 min
3	**Roadmaps to fault tolerance** What companies promise for the next few years.	5 min
4	**Roles in the field** Four common career paths and what each needs.	5 min
5	**Skills and portfolio** Certifications, open source and research internships.	5 min
6	**Further study** Courses, textbooks, communities, competitions and a 12-month plan.	4 min

1

Hardware players

Seven companies, four ways of building a qubit.

Company	Qubit type	Where it stands
IBM	Superconducting	Heron and Nighthawk processors; building toward Starling, a fault-tolerant system targeted for 2029.
Google Quantum AI	Superconducting	Willow chip showed logical errors falling as the error-correcting code grew larger.
IonQ	Trapped ions	Tempo systems; acquired Oxford Ionics; targets 2 million physical qubits by 2030.
Quantinuum	Trapped ions	Helios: 98 qubits with 99.921% two-qubit gate fidelity, among the best reported.
Rigetti	Superconducting	Modular, chiplet-based processors aimed at scaling up the qubit count.
PsiQuantum	Photonic	Makes photonic chips in a chip foundry; plans utility-scale machines in Australia and the US.
QuEra	Neutral atoms	Reconfigurable atom arrays; error-correction experiments with Harvard and MIT.

Based on company announcements and filings up to late 2026; check each company's site for the latest.

2 National missions and funding

Governments treat quantum as a strategic technology and fund it for the long term.

India's National Quantum Mission (NQM)

₹6,003 cr
approved outlay, 2023–24 to 2030–31

50–1,000
physical-qubit computers targeted in 8 years

4 hubs
thematic hubs in top institutes

- **Computing:** IISc Bengaluru
- **Communication:** IIT Madras
- **Sensing and metrology:** IIT Bombay
- **Materials and devices:** IIT Delhi

Hubs work with partner institutions and start-ups.

United States

National Quantum Initiative, in law since 2018, coordinates federal quantum research.

United Kingdom

A ten-year national quantum strategy backed by about £2.5 billion.

European Union

The Quantum Flagship, a 10-year programme of about €1 billion from 2018.

Closer to home: Amaravati Quantum Valley

Andhra Pradesh plans an IBM Quantum System Two with a 156-qubit Heron processor, with TCS and L&T. Commissioning was targeted for 2026.

3

Hardware roadmaps and the road to fault tolerance

Everyone is heading from noisy physical qubits to error-corrected logical qubits.

	2026	2027	2028	2029	2030+
IBM	Kookaburra module; advantage tests	Cockatoo links two modules	15,000-gate circuits	Starling: about 200 logical qubits	Next: 1,000+ logical qubits
Quantinuum	Helios in service: 98 qubits	Sol: about 100 logical qubits	–	Apollo: fault-tolerant target	–
IonQ	Tempo: 100 qubits	10,000 qubits on one chip	20,000 qubits; about 1,600 logical	–	2 million qubits; 40,000–80,000 logical

What is a logical qubit?

Many physical qubits working together under error correction so that one protected qubit survives long computations.

Read roadmaps with care

These are company targets, not delivered results. Dates slip, and “logical qubit” is counted differently by each company.

4

Roles in the quantum industry

Four common career paths; real jobs often blend two of them.

S Quantum software engineer

Does: Builds SDKs, compilers, cloud platforms and error-mitigation tools.

Background: Computer science or software engineering; Python, C++ or Rust; quantum basics.

A Algorithm researcher

Does: Designs and analyses algorithms, error-correcting codes and resource estimates.

Background: Mathematics, computer science or physics, usually with a master's or PhD.

H Hardware engineer

Does: Builds and tests qubits, cryogenics, control electronics, lasers or photonic chips.

Background: Physics or electronics, with microwave, optics or chip-design skills.

P Applications scientist

Does: Turns a chemistry, finance or logistics problem into a quantum workflow and compares it with classical methods.

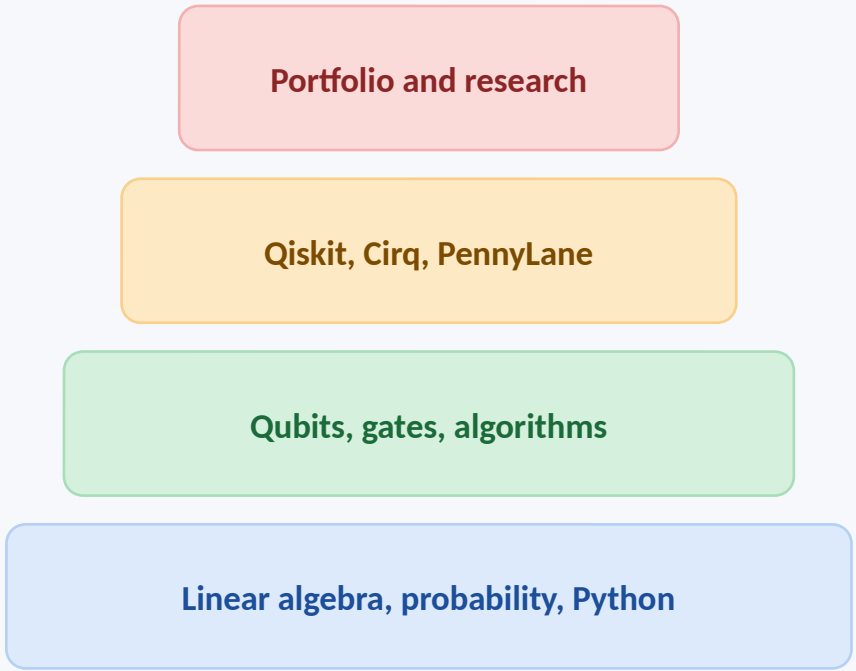
Background: A domain degree plus quantum programming skills.

5

Skills and portfolio

Proof of what you can do counts for more than a list of courses.

The skill stack



Build from the bottom; each layer makes the next one faster.

C

Certifications

IBM offers a Qiskit developer certification that shows you can write and run circuits. Treat it as a starting point, not a substitute for projects.

O

Open-source contributions

Pick Qiskit, Cirq or PennyLane: fix documentation, add a tutorial, then take a “good first issue”. Merged pull requests are public proof.

R

Research internships

Apply to IISc, the IITs, TIFR, NQM hubs and industry labs. Summer research fellowships are a common route in India.

6 Further study: courses, books, communities, contests

Free, high-quality material exists for every stage; the hard part is staying consistent.

M MOOCs and courses

IBM Quantum Learning: free, with Qiskit labs
NPTEL: quantum computing courses from Indian institutes
edX and Coursera: university quantum courses

B Textbooks

Nielsen and Chuang: the standard reference
Kaye, Laflamme and Mosca: a gentler introduction
Preskill's lecture notes: free online

C Communities

Qiskit Slack and the Unitary Fund community
QOSF: open-source mentorship programme
Quantum Computing Stack Exchange and arXiv quant-ph

T Competitions

IBM Quantum challenges and Qiskit Fall Fest
iQuHACK (MIT) and QHack hackathons
Team up: shared projects strengthen a portfolio

Programmes and dates change every year; check their websites for the current schedule.

6

A 12-month plan for a final-year student

Four quarters that turn interest into a portfolio.

Months 1–3

Foundations

- Refresh linear algebra and probability
- Finish the IBM Quantum Learning basics
- Write five small circuits in Qiskit

→

Months 4–6

Build

- Implement Grover and the QFT from scratch
- Run one circuit on real hardware
- Publish it on GitHub with a README

→

Months 7–9

Contribute

- Fix documentation or a small bug in an open-source library
- Join a community or mentorship programme
- Enter one hackathon

→

Months 10–12

Specialise

- Pick a role: software, research, hardware or applications
- Apply for an internship or fellowship
- Draft a one-page portfolio

Keep a public log: one GitHub repository per project and a short write-up of what you learned. It becomes your portfolio.

ONE-LINE TAKEAWAY

The quantum industry is growing fast and has room for many roles; a public portfolio of projects, open-source work and research will show your skills better than any single certificate.

Many routes to fault tolerance

India: NQM and Amaravati

Portfolio beats credentials

→

Next: Course wrap-up and revision

We pull the whole course together and plan how to keep learning.